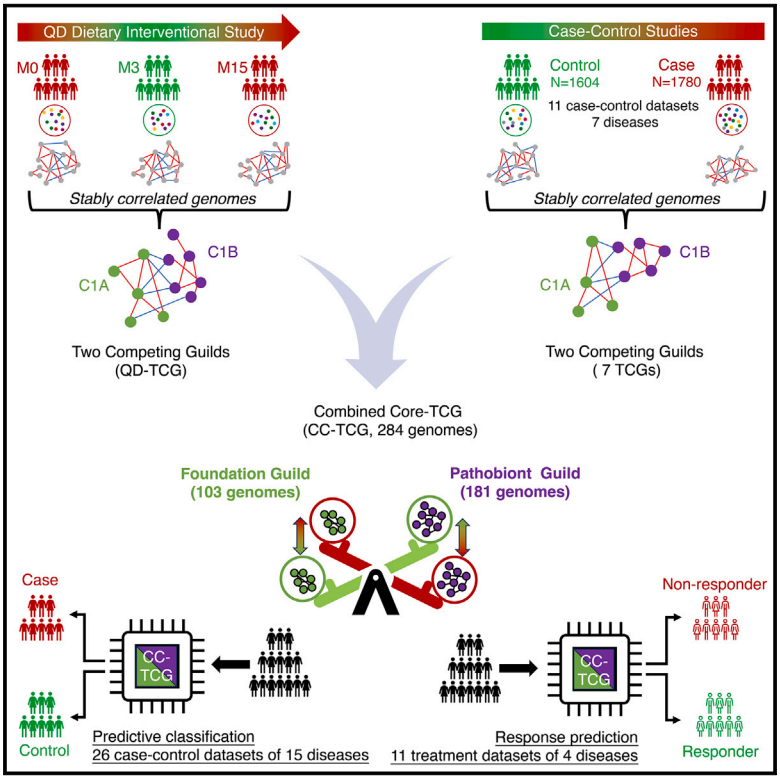
Cell

A core microbiome signature as an indicator of health

Graphical abstract



Highlights

- Integrates diverse microbiome datasets within the same analytical framework
- Uncovers stable relationships among gut bacteria across interventions and disease states
- Reveals a core structure of two competing guilds across populations and health status
- Offers stably interacting microbial guilds as novel targets for health modulation

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In brief

This study identifies gut bacteria that maintain stable, enduring relationships under various perturbations, such as dietary changes and disease states. These bacteria form a core microbiome structure of two competing guilds (TCGs), one beneficial, the other potentially detrimental, across populations and health statuses, offering a novel target for modulating human health.

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A core microbiome signature as an indicator of health

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SUMMARY

The gut microbiota is crucial for human health, functioning as a complex adaptive system akin to a vital organ. To identify core health-relevant gut microbes, we followed the systems biology tenet that stable relationships signify core components. By analyzing metagenomic datasets from a high-fiber dietary intervention in type 2 diabetes and 26 case-control studies across 15 diseases, we identified a set of stably correlated genome pairs within co-abundance networks perturbed by dietary interventions and diseases. These genomes formed a "two competing guilds" (TCGs) model, with one guild specialized in fiber fermentation and butyrate production and the other characterized by virulence and antibiotic resistance. Our random forest models successfully distinguished cases from controls across multiple diseases and predicted immunotherapy outcomes through the use of these genomes. Our guild-based approach, which is genome specific, database independent, and interaction focused, identifies a core microbiome signature that serves as a holistic health indicator and a potential common target for health enhancement.

INTRODUCTION

The gut microbiome, akin to an essential organ, plays a pivotal role in host health, influencing processes ranging from nutrient metabolism to immune function and behavior.[1,2] Advances in high-throughput sequencing technology have extensively characterized this complex ecosystem, unveiling detailed insights into its composition and functionality. However, conventional analytical methods, such as taxonomy-based and gene-centric approaches, often yield inconsistent results. For instance, taxonomic groups like Lactobacillus or Firmicutes have shown positive, negative, or no correlation with diseases like obesity or diabetes across different studies.[3] Consequently, there is a pressing need for a new methodological framework to robustly identify key microbiome members as biomarkers for health maintenance and disease management.

Traditional methods often rely on low-resolution features like taxonomic tags, which fail to discern subtle genetic and functional variations within the same taxon in microbial communities. To enhance the resolution of microbial analysis, we advocate using *de novo* assembled high-quality metagenome-assembled genomes (HQMAGs).[4] HQMAGs enable near-strain-level resolution by employing a 1% average nucleotide identity (ANI) differentiation threshold—significantly finer than the typical 5%–6% used for species definition.[5] Moreover, to facilitate database-independent tracking of microbial entities across various studies and samples, we assign each HQMAG a universal unique identifier (UUID). This strategy includes novel or understudied

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microbes in the analysis and enhances data comparison and integration across different projects by following these UUIDs as genome tags. HQMAGs provide a deeper understanding of the functions of health-relevant microbes by analyzing all their genes within their genomic context, enabling functional analysis even for unannotated genes. This strategic refinement in microbial genome analysis sets the stage for more precise identification and understanding of microbial roles in health and disease.

Furthermore, traditional analytical methods often focus on differential analysis of single features, disregarding the complex interactions among microbes. Such practices overlook the reality that the gut microbiome exemplifies a complex adaptive system (CAS) characterized by components or agents that interact in dynamic and non-linear ways.[6] These interactions lead to system-level emergent properties that cannot be predicted from the behavior of individual agents alone. Within this microbiome CAS, microbes form structural modules known as ''guilds,'' underpinning the emergent properties of the gut ecosystem that support host health.[7] Members of the same guild may come from different taxonomic backgrounds but thrive and decline together, exhibiting co-abundance behavior. We employ co-abundance analysis of HQMAGs to delineate guild-level organization based on interactions among microbes, illuminating ''who works with whom'' within this intricate ecosystem. Through this lens, each guild is correlated with human clinical parameters, providing insights into their impact on health and disease and embodying the CAS principle that ''the whole is greater than the sum of its parts.''

Using this innovative methodology, our previous works have elucidated functional groups associated with conditions including obesity,[8] type 2 diabetes (T2D),[9] diabetic kidney disease,[10] diabetic neuropathy,[11] and coronavirus disease 2019 (COVID-19).[12] These studies underscore microbial interactions' critical role in the microbiome's functional dynamics and their broader implications for host health. Hence, this study adopts a guild-based framework emphasizing genome specificity, database independence, and an interaction-focused analysis. This approach shifts how we view microbial entities—from considering them independent units to exploring their intricate interaction patterns. Such a perspective is crucial for probing the system-level functions of the gut microbiome and understanding their impact on host health. This foundational understanding guides our exploration into how these complex microbial networks influence overall health outcomes.

Leveraging our guild-based approach, this study concentrates on stable interactions to pinpoint health-relevant microbial entities. Inspired by the systems biology tenet that stable interactions signify core system components,[13] we hypothesize that stably correlated genome pairs across diverse conditions are likely core members of health-relevant guilds, potentially shaped by evolutionary forces due to their critical roles in human health. By constructing co-abundance networks of HQMAGs from individuals under various environmental perturbations, including dietary interventions and health state differences, we have uncovered a core set of stably correlated HQMAGs. These HQMAGs form a robust network of two competing guilds (TCGs) and have been employed as core features in developing machine learning models. These models differentiate between cases and controls across 26 datasets involving 15 different diseases and predict responses to immunotherapy in four distinct conditions, achieving moderate to excellent performance. This TCG model represents a core microbiome structure of key health-relevant guilds driven by stable microbial interactions. Such findings could substantially advance precision microbiome medicine by leveraging the stability of microbial interactions as biomarkers to improve disease diagnosis and management.

RESULTS

Stably correlated genome pairs under high-fiber perturbations constitute a subnetwork of TCGs

To identify stably correlated bacterial genomes, we focused on a gut microbiota-targeted high-fiber interventional trial called QD. In this trial, T2D patients were randomized at baseline (M0) to receive either 3 months (M3) of a high-fiber intervention (W group; n = 74) or standard care (U group; n = 36), followed by a 1-year follow-up (M15) (Figures 1 and 2A; Table S1; Data S1, page 1). The QD trial examined the gut microbiota's response to dietary alterations in T2D patients (Figure 2B; Data S1, page 2). Through shotgun metagenomic sequencing, 1,845 non-redundant HQMAGs were *de novo* reconstructed, representing over 70% of the total reads from 315 fecal samples collected at M0, M3, and M15.

A significant alteration in the gut microbiota structure was observed in the high-fiber diet group from M0 to M3, which reverted to baseline at M15, indicating high resilience in community structure, though there may be a lingering increase in richness (Figures 2C and 2D; Data S1 pages 3 and 4). This reversion in microbiota structure mirrored the reversible changes observed in host metabolic phenotypes, such as hemoglobin A1c levels (Figures 2E–2H; Table S2). These dramatic gut microbiota structural changes in the W group patients provided an opportunity to identify stably correlated genomes as potential health-relevant microbial members.

A total of 477 HQMAGs were selected as prevalent and predominant members of the gut microbiome in the W group, as they were detectable in more than 75% of the samples at each time point and accounted for ~60% of the total abundance of the 1,845 HQMAGs. We calculated pairwise correlations among these 477 HQMAGs based on their abundance across the patients at each time point and ultimately obtained three scale-free co-abundance networks (G_{M0}, G_{M3}, and G_{M15}) (Figure 3A; Data S1 pages 5 and 6) using FastSpar.[14] Compared with G_{M0}, G_{M3} showed a significantly changed degree distribution and a loss of network hubs, which rebounded at G_{M15}, indicating a profound change in the microbiome's overall structure during the trial (Data S1, page 7).

Amid these dramatic environmentally induced changes during the dietary intervention, we searched for stably correlated genomes indicative of resilient ecological relationships. Out of the 113,526 possible genome pairs, a vast majority (92.39%) exhibited no significant correlations at any time point, suggesting that ecological relationships among bacteria are rare (Figure 3B). Yet, 635 stable correlations involving 184 HQMAGs were identified via checking the edge vectors through summation, showing the same direction of correlation between the identical genome

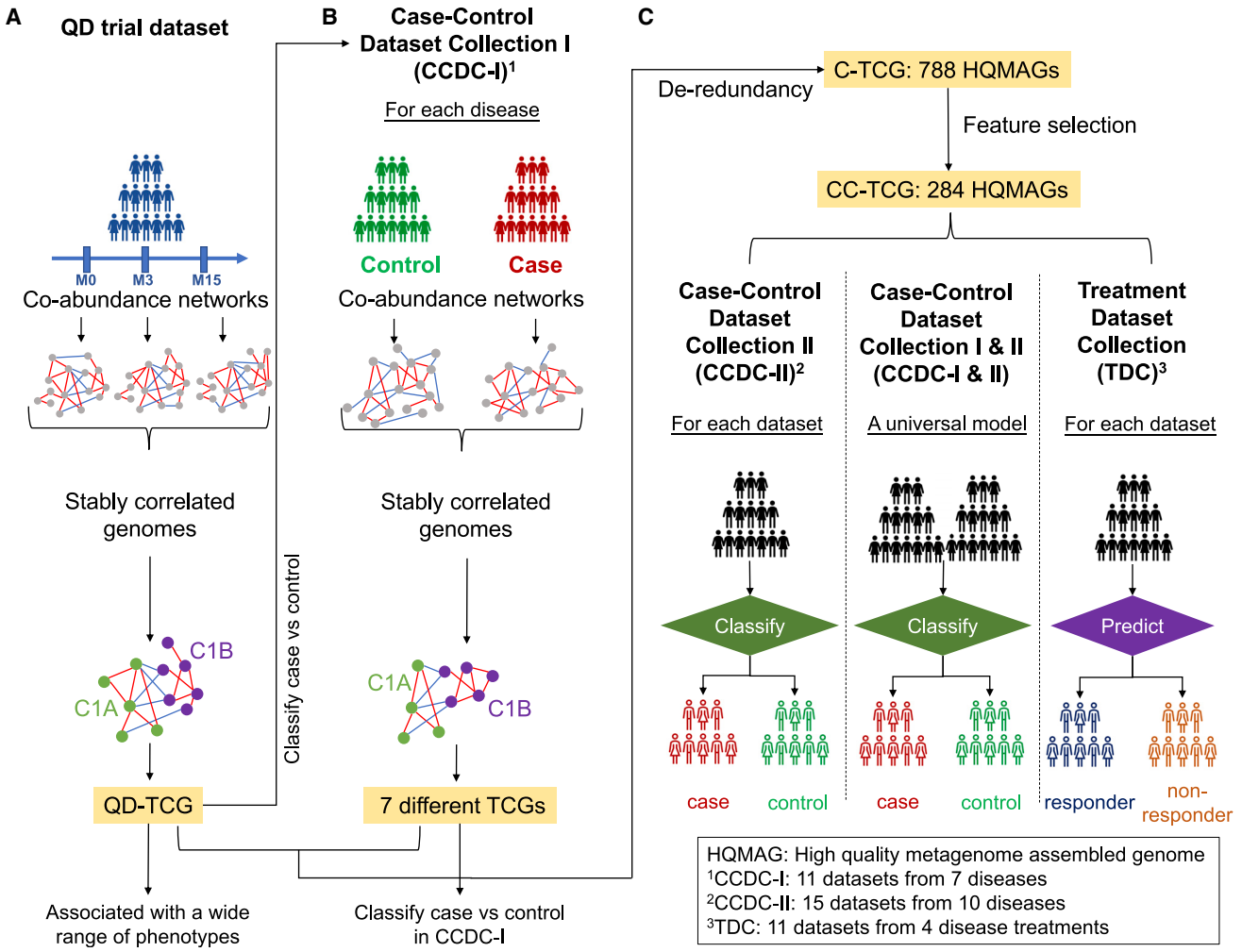


Figure 1. Study scheme and datasets employed

(A and B) Identification of two competing guilds (TCGs) with stable relationships in the (A) QD trial and (B) case-control dataset collection I (CCDC-I).

(C) Integrating TCGs found in (A) and (B) via genome-level dereplication (99% ANI) and performing feature selection to find core members of TCGs as CC-TCG. CC-TCG was further used to classify case vs. control in case-control dataset collection II (CCDC-II). Subsequently, a universal model was built on CC-TCG to discriminate cases from controls across all samples in CCDC-I and CCDC-II. In addition, CC-TCG was used to predict treatment responses based on pre-treatment gut metagenomes in the treatment dataset collection (TDC).

See also Figures S1 and S2.

pairs across all three time points (see STAR Methods). Connected components clustering analysis organized these stably correlated genomes into 17 distinct unconnected clusters (Figure 3C). Cluster C1 had 141 HQMAGs with both positive and negative correlations, whereas clusters C2–C17 held 2–9 HQMAGs with only positive correlations. Weighted correlation network analysis (WGCNA)[15] based on average linkage tree built upon correlations further divided C1 into two sub-clusters, C1A with 50 HQMAGs and C1B with 91, which notably were the only clusters significantly correlated with the primary outcome HbA1c (Figure 3D), thereby becoming the focus for subsequent analysis.

C1A and C1B are potential guilds since HQMAGs within each cluster exhibit co-abundance behavior through only positive correlations, indicating a potential cooperative relationship (Figure 3E; Data S1, page 8). These two guilds are exclusively linked through negative edges, showcasing a potential competitive relationship, a feature termed TCGs identified from the QD trial dataset (QD-TCG). The genomes in QD-TCG notably possess significantly higher degrees, betweenness centrality, eigenvector centrality, closeness centrality, and stress centrality than other genomes in the entire network (Data S1, page 9). This highlights that the two guilds wield substantial control over other nodes' interactions and the network's information flow. Removing QD-TCG would precipitate a network collapse since an average of 86.08% of the total edges would be lost (Data S1, page 9). This underscores that the genomes within QD-TCG can be recognized as the core nodes of G_{M0}, G_{M3}, and G_{M15}, owing to their stable and high connections, making them the backbone of the gut ecosystem network in patients of the W group.

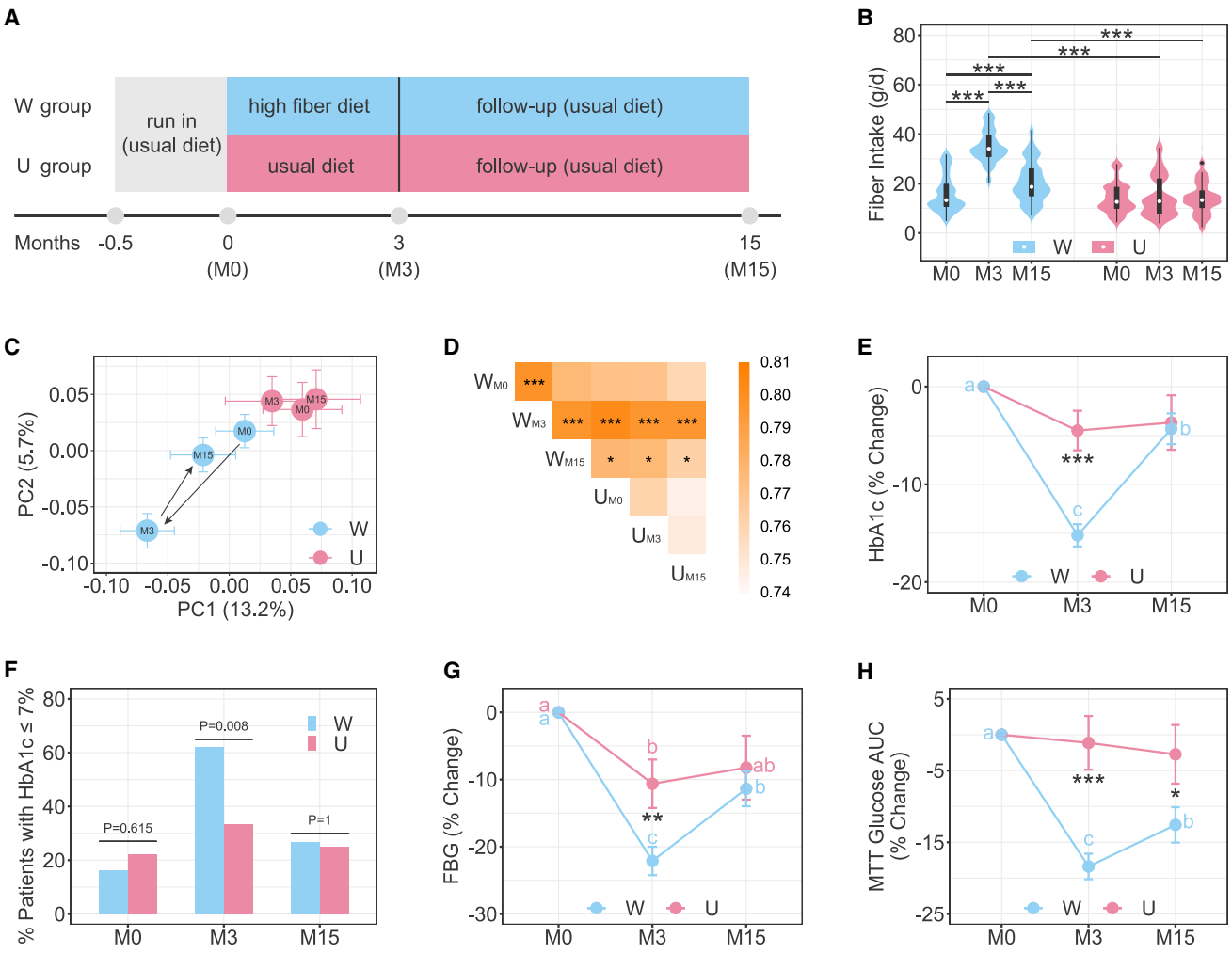


Figure 2. Reversible alterations in the gut microbiota induced by a high-fiber diet are associated with corresponding shifts in metabolic phenotypes in patients with T2D

(A) Study design of the QD trial. After the run-in, medical checkup and sample collection were conducted at baseline (M0), 3 months (M3) after the high-fiber intervention (W) or standard care with usual diet (U), and 1 year (M15) after the intervention stopped.

(B) Changes in fiber intake.

(C and D) (C) Global changes of the gut microbiome as shown by the principal coordinate analysis based on the Bray-Curtis distance with the 1,845 HQMAGs and (D) average Bray-Curtis distance between the groups. PERMANOVA test (9,999 permutations) was performed to compare the groups. * $p < 0.05$ and *** $p < 0.001$. The color intensity of the square showed the magnitude of the average Bray-Curtis distance.

(E–H) (E) Change of HbA1c, (F) the percentage of participants who achieved adequate glycemic control, (G) fasting blood glucose, and (H) the glucose area under the curve (AUC) in a meal tolerance test (MTT). For (E), (G), and (H), data were shown as percent changes from baseline (± SEM). Friedman test, followed by the Nemenyi post hoc test, was used for comparison in the same group; compact letters reflect significance ($p < 0.05$). $n = 67$ in W group and $n = 28$ in U group. Mann-Whitney test (two-sided) was used for comparison between W and U at the same time point, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. $n = 74$ in W (M0) (for H, $n = 72$), $n = 74$ in W (M3), $n = 67$ in W (M15), $n = 36$ in U (M0), $n = 36$ in U (M3), and $n = 28$ in U (M15).

All members in C1A were from the phylum Firmicutes, whereas those in C1B were from five different phyla, including Firmicutes, Bacteroidota, Proteobacteria, Actinobacteriota, and Fusobacteriota. HQMAGs in QD-TCG were prevalent, with 137 found in over 90% of individuals (Data S1, page 10). They were also predominant, with 111 surpassing the median abundance of the 1,845 HQMAGs, accounting for 20.78% of the total sequencing reads. Moreover, the variations within QD-TCG captured the major variations of beta diversity observed in the entire dataset based on all HQMAGs (Data S1, page 11).

Next, we examined whether and how QD-TCG responded to the high-fiber intervention. In the W group, the abundance of C1A increased, and C1B decreased significantly from M0 to M3. Then, at M15, C1A decreased to a level similar to that at M0, and C1B increased but was not significantly different from that at M3. Subsequently, from M0 to M3, high-fiber intervention significantly increased the C1A-to-C1B ratio. At one-year follow-up, this ratio significantly decreased from M3 (Figure 4A). Neither the abundances of the two guilds nor their ratio changed in the U group across the trial (Data S1, page 12). Thus, the changes in

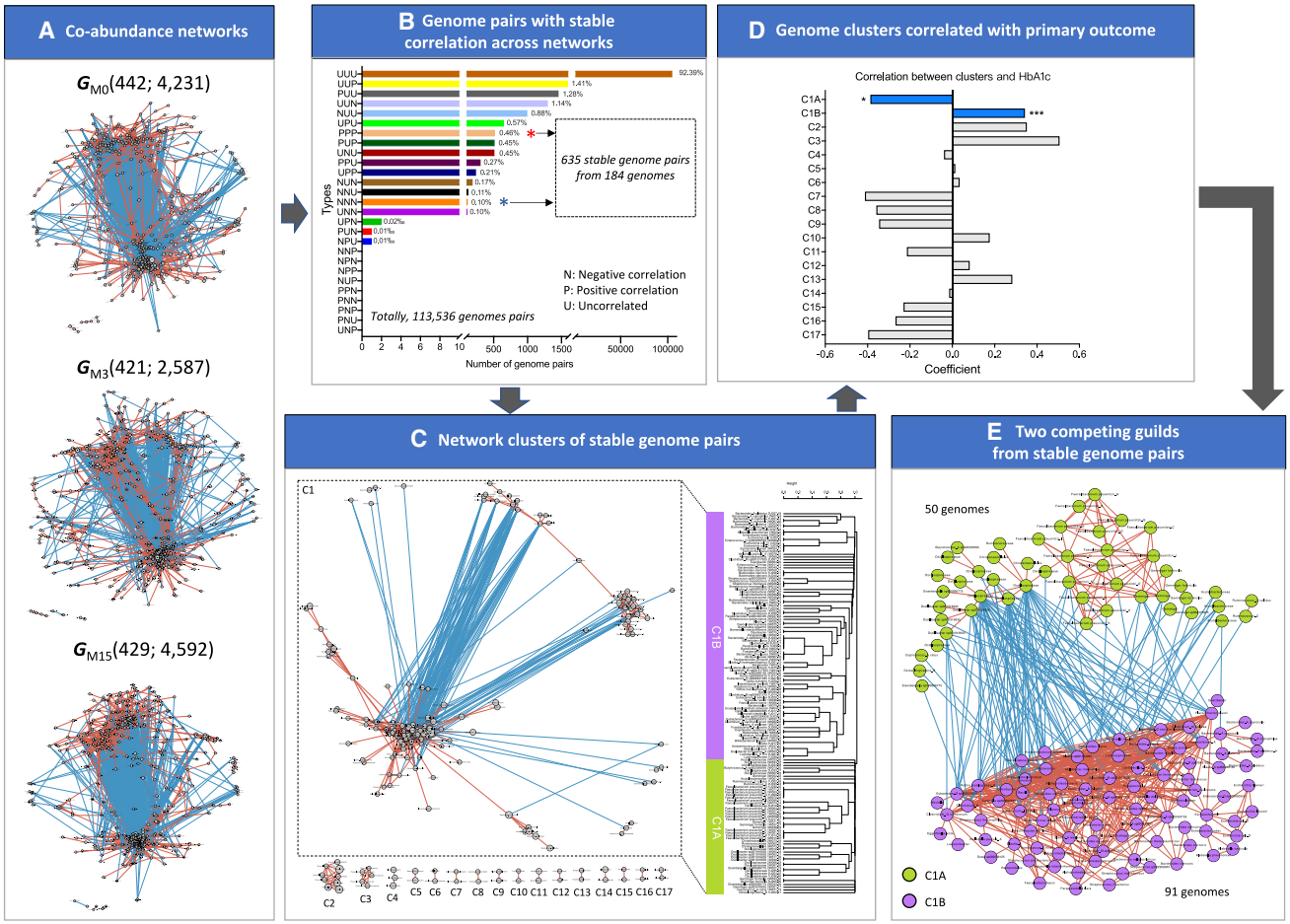


Figure 3. Despite substantial global changes in the gut microbiota induced by the high-fiber intervention, stable genome pairs form TCGs (QD-TCG) associated with HbA1c levels

(A) Co-abundance networks of the prevalent 477 HQMAGs in the W group at M0, M3, and M15 during the trial, denoted as G_{M0}(442; 4,231), G_{M3}(421; 2,587), and G_{M15}(429; 4,592), and numbers in parenthesis are order and size of the network. The correlations between HQMAGs were calculated using FastSpar, n = 67 patients. All significant correlations with $p \leq 0.001$ were included. Edges between nodes represent correlations. Red and blue colors indicate positive and negative correlations, respectively. Node size indicates the average abundance of the HQMAGs. Edge-weighted spring embedded layout determined the layout of the nodes and edges with correlation efficient as weight.

(B) The distribution of different types of correlations of the genome pairs during the trial. The three letters show the correlations (N for negative, P for positive, and U for uncorrelated) of the genome pairs at M0, M3, and M15. Stable correlations across time points, NNN and PPP, were indicated by asterisks.

(C) HQMAGs with stable correlations were clustered into 17 clusters (C1–C17) by using connected components clustering analysis. C1 was further clustered into C1A and C1B by using average linkage and WGCNA analysis based on negative and positive correlations.

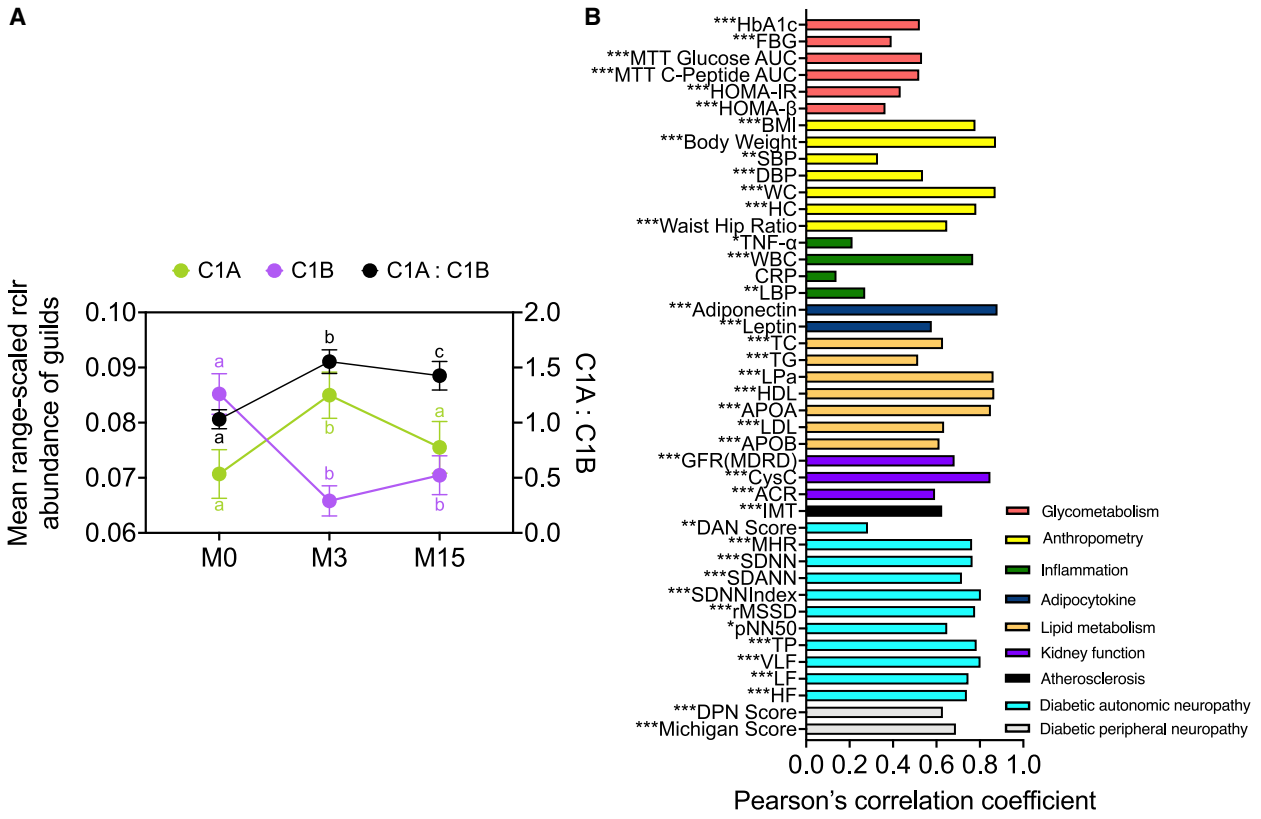
(D) Correlations between genome clusters and HbA1c using linear mixed effect model by MaAslin2 package. Abundance was log-transformed. The subject was used as a random effect. n = 67. p values were adjusted using Benjamini & Hochberg's method. * BH adjusted $p < 0.05$, *** BH adjusted $p < 0.001$.

(E) The network of QD-TCG.

the two guilds were concomitant with changes in dietary fiber intake, overall gut microbiota, and host phenotypes, particularly an increase of C1A and a decrease of C1B, which were associated with improved host phenotypes. This suggests that C1A is potentially beneficial and C1B detrimental.

To reveal whether QD-TCG is associated with other host metabolic phenotypes in addition to the primary outcome, we applied a linear mixed-effect model to identify the associations between QD-TCG and host bio-clinical parameters. We used the abundances of C1A and C1B and bio-clinical parameters at M0 and M3 for training models to predict bio-clinical parameters with the abundances of C1A and C1B at M15. Forty-two out of the 43 bio-clinical parameters had a significant correlation between predicted values and actual values, with the Pearson's coefficient ranging from 0.14 to 0.88 (adjusted p value < 0.05) (Figure 4B). These results showed that QD-TCG constitutes an important microbiome signature significantly associated with a wide range of metabolic phenotypes in T2D.

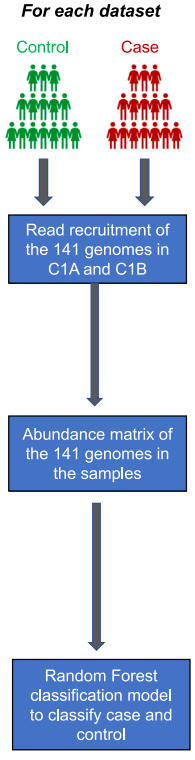
Next, we explored the potential of QD-TCG as a biomarker to differentiate T2D from control subjects in an independent metagenomic dataset comprising 136 T2D cases and 136 controls.[16] We estimated the abundances of these 141 HQMAGs from



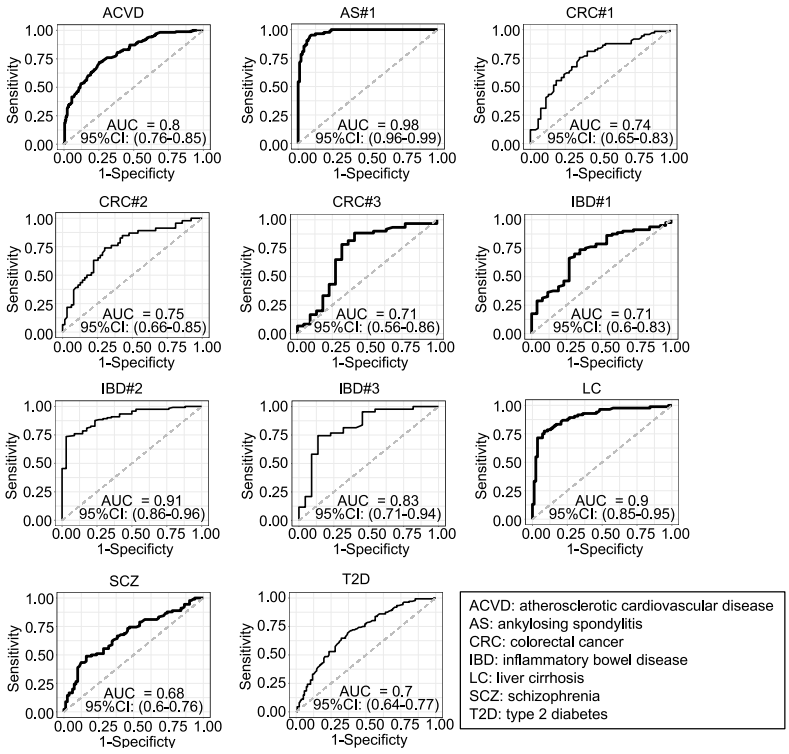
C

Case-Control Dataset Collection I

ACVD	China	
214	171	ACVD
AS	China	
83	97	AS#1
CRC	China	
52	73	CRC#1
CRC	Austria	
63	46	CRC#2
CRC	Germany	
60	21	CRC#3
IBD	USA	
26	80	IBD#1
IBD	USA	
34	121	IBD#2
IBD	Netherlands	
22	43	IBD#3
LC	China	
83	84	LC
SCZ	China	
81	90	SCZ
T2D	China	
136	136	T2D



D



(legend on next page)

QD-TCG via a read recruitment analysis to develop a random forest classifier to distinguish cases from controls. Receiver operating characteristic curve analysis demonstrated a moderate diagnostic capability, with an area under the curve (AUC) of 0.70, ascertained through leave-one-out cross-validation (Figures 4C and 4D).

To assess if QD-TCG might represent an intrinsic pattern in the human gut microbiome, irrespective of disease types, we collected ten more independent metagenomic datasets of cases and controls spanning six different diseases: atherosclerotic cardiovascular disease (ACVD),[17] ankylosing spondylitis (AS, AS#1),[18] colorectal cancer (CRC, CRC#1, #2, and #3),[19–21] inflammatory bowel disease (IBD, IBD#1, #2, and #3),[22,23] liver cirrhosis (LC),[24] and schizophrenia (SCZ),[25] collectively referred as the case-control dataset collection I (CCDC-I). On average, in each dataset of CCDC-I, 32.72% and 31.41% reads were recruited to QD-TCG in the case and control samples, respectively (Figure 4C). Within each dataset, a random forest classifier using the 141 HQMAGs distinguished cases from controls, with the AUC ranging from 0.68 for SCZ to 0.98 for AS#1 (Figure 4D). This suggests that QD-TCG might represent a common health-relevant microbiome signature associated with a broad spectrum of human diseases.

Stably correlated genome pairs under disease-induced perturbations constitute similar subnetworks of TCGs

To explore whether TCGs could also be identified in the context of disease-induced perturbations, we applied our methodology to case-control datasets from CCDC-I (Figure 5A). We first reconstructed 682, 384, 564, 1,308, 812, 1,644, and 774 *de novo* HQMAGs in the case and control datasets for T2D, LC, AS#1, ACVD, SCZ, CRC, and IBD, respectively. We then built co-abundance networks for case and control groups using HQMAGs present in over 75% of subjects in each group to identify stably correlated genomes (Data S2).

Within each disease type, we identified a primary cluster (C1) encompassing most of the stably networked genomes with both positive and negative correlations. These clusters were further divided into two sub-clusters using WGCNA analysis (Data S2). Sub-clusters with greater average ANI alignment to QD-TCG's C1B were labeled C1B, while the others were labeled C1A. Most stable correlations among HQMAGs within C1A and C1B were positive, while correlations between C1A and C1B were predominantly negative (Figures 5B–5H). This pattern confirmed the presence of TCGs in the gut microbiome across various diseases, ethnicities, and geographical regions.

To evaluate the diagnostic potential of each TCG identified from CCDC-I, we conducted a read recruitment analysis on its HQMAGs across all other metagenomic datasets in CCDC-I. Using these data, we developed random forest classifiers to distinguish between cases and controls. In most datasets, TCGs from CCDC-I exhibited moderate to excellent diagnostic power for differentiating cases from controls in their original and independent datasets (Figure 5I; Data S2). These findings suggest that the TCGs identified in case-control datasets represent a consistent health-relevant microbiome signature associated with a wide range of human diseases.

Integrated TCGs exhibit enhanced performance in classifying cases from controls and are validated in more independent datasets

Given that both QD-TCG and the seven TCGs from CCDC-I successfully distinguished cases from controls across various diseases, we aimed to integrate these eight TCGs to develop a unified set of predictors. After performing a de-redundancy analysis based on a genomic ANI cutoff of >99%, we identified a pool of 788 non-redundant HQMAGs, referred to as the combined TCG (C-TCG) (Figures 1 and S1). Using the abundance matrix of these 788 HQMAGs, we trained a random forest classifier on each CCDC-I dataset (Figure S2A). The C-TCG demonstrated moderate to excellent diagnostic power for distinguishing cases and controls within CCDC-I (Figure S2B).

Next, we sought to identify the most relevant C-TCG members for classification performance as the core genome predictors. Through a feature importance-based and cross-model ranking method (see STAR Methods), we found that classifiers trained on the top 302 HQMAGs exhibited the best classification performance in CCDC-I, as indicated by the smallest cumulative rank (Figure S1). Among these 302 HQMAGs, 103 were unique to C1A, 181 to C1B, and 18 showed inconsistent assignments across different TCGs. Discarding the 18 inconsistent HQMAGs resulted in a set of 284 HQMAGs consistently assigned to the TCGs, referred to as the combined core set of the TCGs (CC-TCG). Random forest classifiers built on CC-TCG demonstrated superior performance in classifying cases and controls compared with those based on C-TCG, QD-TCG, and individual TCGs from CCDC-I (Figures S2C–S2E). This

Figure 4. Genomes within the TCGs (QD-TCG) predict metabolic health outcomes in T2D patients of the QD trial and distinguish cases from controls in CCDC-I

(A) Change of the mean range-scaled robust centered-log-ratio (rclr) transformed abundances of C1A, C1B, and their ratio across the trial in the W group. Friedman test, followed by the Nemenyi test, was used to analyze the difference between time points. Compact letters reflect the significance at $p < 0.05$. Data are presented as the mean ± SEM.

(B) Prediction of bio-clinical parameters with the genomes in the TCGs. Linear mixed effect models were trained based on the mean range-scaled rclr abundances of C1A and C1B and clinical parameters at M0 and M3 with the subject as a random effect. The mean range-scaled rclr abundances of C1A and C1B at M15 were used to predict clinical parameters based on the trained model. The bar plot shows Pearson's correlation coefficient between the predicted clinical parameters and measured values at M15. The asterisk before the parameter's name indicates significant Pearson's correlations. p values were adjusted using Benjamini & Hochberg's method. * adjusted $p < 0.05$, ** adjusted $p < 0.01$ and *** adjusted $p < 0.001$.

(C) Datasets from 11 different datasets on seven diseases were collected as case-control dataset collection I. Sample size of each case-control dataset is shown in red and green numbers in the left panel, respectively. For each dataset in the collection, metagenomic reads were recruited to the 141 genomes of the QD-TCG to estimate genome abundance in each sample.

(D) Based on the abundance matrix of the 141 genomes, a random forest classification model with leave-one-out cross-validation was trained to classify case and control subjects in each dataset. The area under the receiver operating characteristic (AUROC) is shown here.

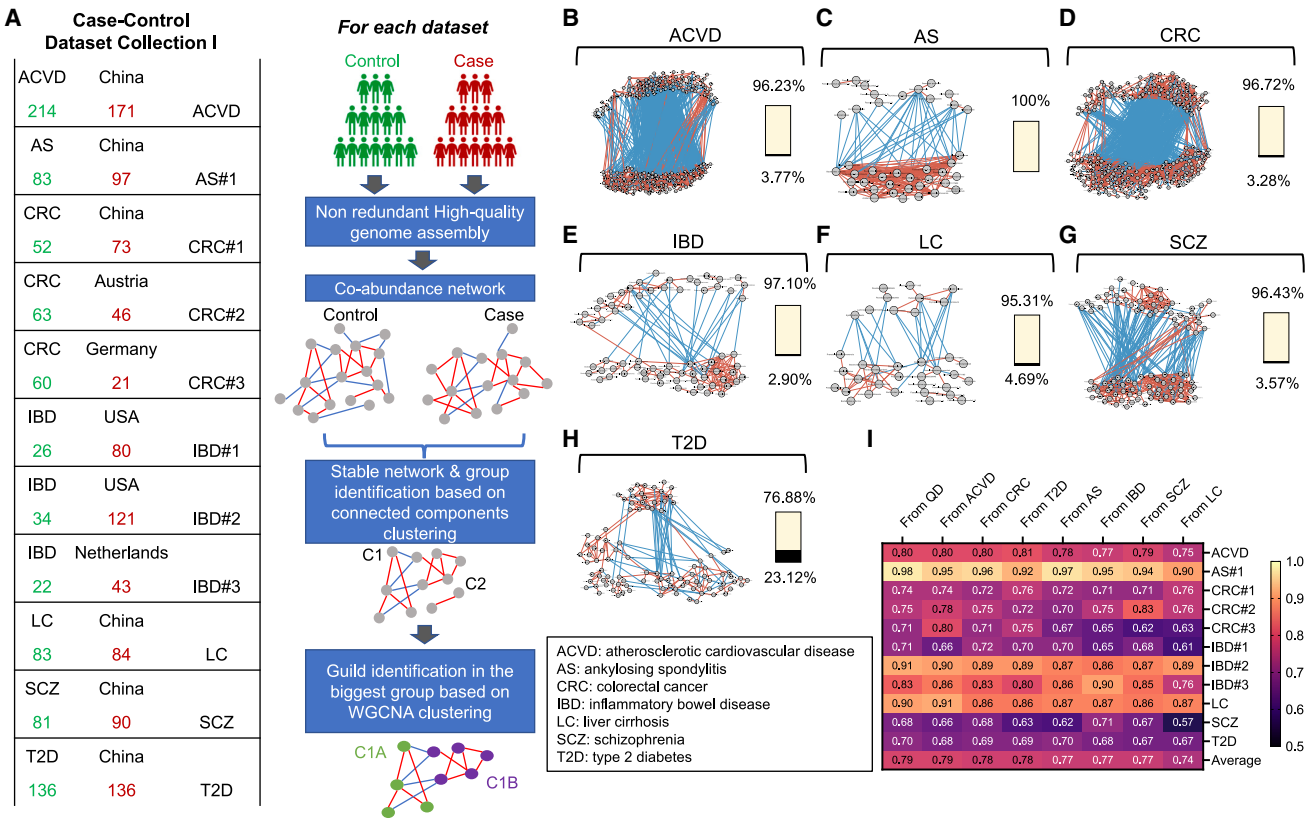


Figure 5. Genomes forming the TCG, identified from datasets specific to one disease, demonstrate significant effectiveness in classifying cases from controls across independent datasets on different diseases within the CCDC-I

(A) Identification of the TCGs in CCDC-I. The sample size of each case-control study is shown in red and green numbers in the left panel, respectively. CCDC-I has 11 published metagenomic case-control datasets on seven diseases. Three CRC datasets were combined to analyze CRC. Three IBD datasets were combined to analyze IBD. The percentage of correlations that followed the pattern of TCGs (i.e., positive edges within each guild, negative edges between the two guilds) was in yellow. The ratio of contradictory correlations, i.e., those negative within each guild and positive between the guilds, was in black of the 100% stacked bar. (B–H) TCGs found in CCDC-I.

(I) The TCG found in one dataset was used as a predictor to classify cases and controls in the same dataset and all other ten datasets. A random forest classification model with leave-one-out cross-validation was applied in each dataset on each set of the TCGs. The area under the ROC curve (AUC) values were shown in the heatmap.

finding suggests that the 284 HQMAGs in CC-TCG constitute a core set of genome predictors highly relevant to human health.

To further validate CC-TCG's capability in distinguishing cases from controls, we compiled an additional 15 independent metagenomic datasets from 10 different diseases, collectively referred to as the case-control dataset collection II (CCDC-II) (Figure 6A). Using the 284 HQMAGs in CC-TCG as reference genomes, we recruited 34.41% and 34.76% of reads from case and control samples, respectively, in CCDC-II. For each dataset in CCDC-II, we trained a random forest classifier using the abundance of these 284 HQMAGs to distinguish cases from controls. The CC-TCG showed moderate to excellent diagnostic power in 10 of the 15 datasets, including AS,[26] autism spectrum disorder (ASD),[27] COVID-19,[28] CRC,[29] Graves's disease (GD),[30] hypertension (HT),[31] multiple sclerosis (MS),[32] and pancreatic cancer (PC).[33,34] For the remaining five datasets, it achieved an AUC value of 0.58 for HT#2[35] and AUC values between 0.6 and 0.7 for Behcet's disease (BD),[36] Parkinson's disease (PD),[37] CRC#4,[38] and CRC#5[39] datasets (Figure 6B).

For diseases like CRC, IBD, and PC, which had multiple datasets, we performed cross-dataset analysis and leave-one-dataset-out (LODO) analysis to assess the general applicability of CC-TCG's diagnostic power. While the model's transportability was insufficient for CRC, it showed moderate to outstanding performance in the IBD and PC datasets. Pooling training datasets in LODO analysis improved CC-TCG-based models' performance for CRC, with AUC values ranging from 0.66 to 0.79. For IBD and PC, LODO analysis produced AUC values ranging from 0.75 to 0.91 and 0.71 to 0.72, respectively (Figure S3). These results validate the classification power of CC-TCG across independent datasets.

We then investigated whether a universal model based on CC-TCG could distinguish cases from controls regardless of disease type. We combined all 26 datasets from CCDC-I and CCDC-II, totaling 1,780 cases and 1,604 controls, and randomly allocated samples for training (80%) and testing (20%) a random forest classifier based on CC-TCG (Figure 6C). Using 10-fold cross-validation, the universal classifier displayed moderate ability,

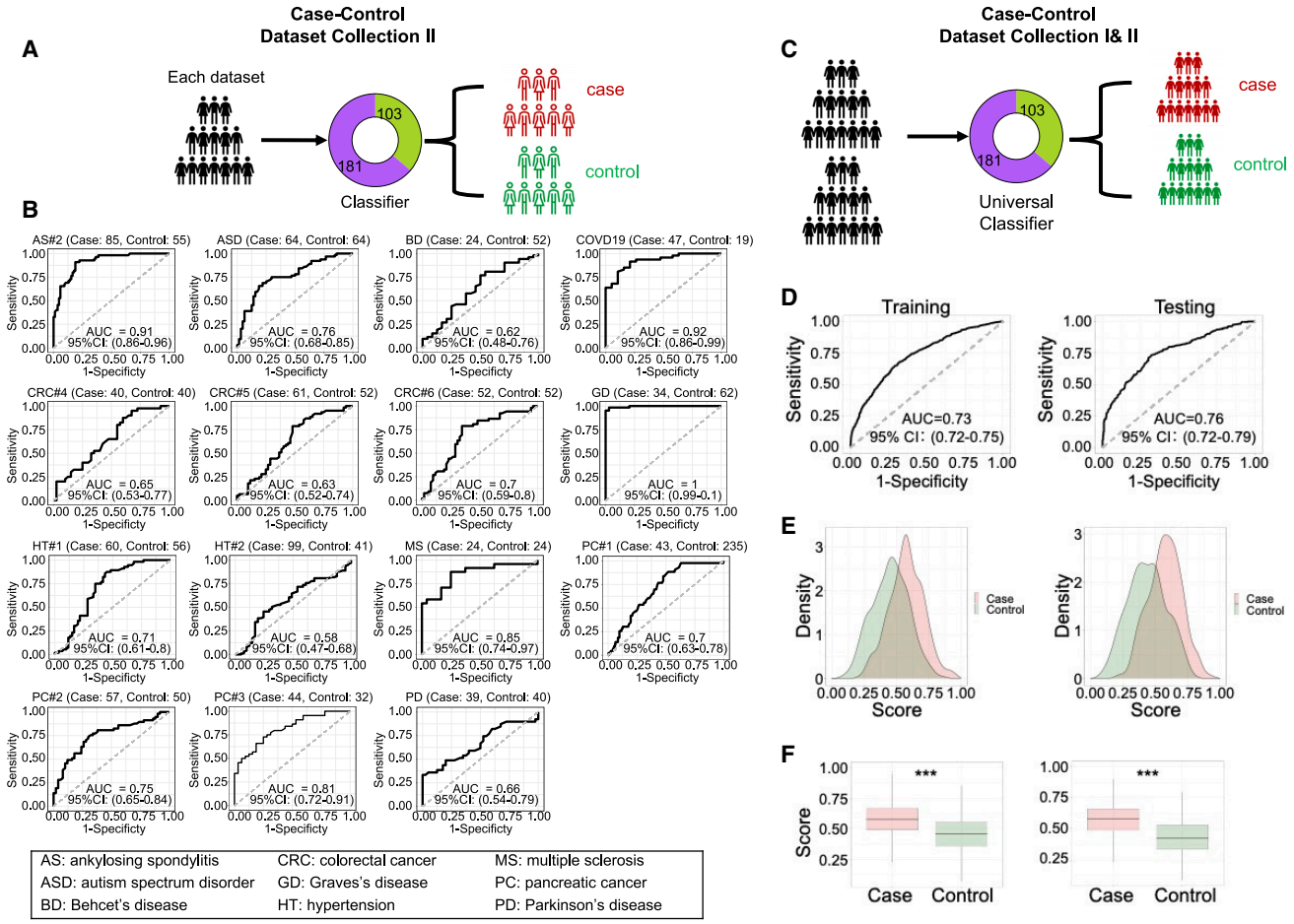


Figure 6. The CC-TCG effectively differentiates cases from controls across a broader range of diseases in CCDC-II and can build a universal model to classify cases and controls across diseases in CCDC-I and CCDC-II

(A) CC-TCG was used as predictors in case-control dataset collection II, which has 15 published metagenomic case-control datasets on ten diseases. A random forest classification model with leave-one-out cross-validation was applied to each dataset.

(B) ROC curve and the area under the ROC curve (AUC).

(C) All control and case samples from case-control dataset collection I and II, encompassing 26 datasets on 15 different diseases, were combined and randomly allocated, with 80% used for training a random forest classification model and 20% for testing.

(D) ROC curve and the area under the ROC curve (AUC).

(E) The density plot of the probability score between case and control. The probability score was generated from the random forest classification model and showed the probability of one sample to be predicted as a case.

(F) Box plot of the probability score between control and case samples. Mann-Whitney test was applied. *** $p < 0.001$. Training: control $n = 1,285$, case $n = 1,424$; testing: control $n = 319$, case $n = 356$.

See also Figure S3.

with an AUC of 0.73, to distinguish cases from controls (Figure 6D). The model's probability score, indicating the likelihood of a sample being classified as a case, showed a clear separation between peaks in both case and control samples (Figure 6E). Controls had significantly lower probability scores than cases (Figure 6F). Applying the universal classifier to the testing data achieved an AUC of 0.76 (Figure 6D). In the testing data, controls also had significantly lower probability scores than cases (Figures 6E and 6F). For both the training and testing data, a probability score of 0.52 provided classification specificity and sensitivity of 0.7 for distinguishing cases from controls. These results indicate the feasibility of using CC-TCG to differentiate cases from controls with a universal model, suggesting that variations in CC-TCG may serve as a generalized indicator of human health.

Integrated TCGs predict immunotherapy responses in independent treatment datasets

Previous reports have linked the gut microbiota to the efficacy of biological therapies in various diseases. We hypothesized that pre-treatment variations in CC-TCG might predict clinical success in these treatments. To test this hypothesis, we compiled 11 pre-treatment metagenomic datasets related to four diseases: IBD,[40,41] rheumatoid arthritis (RA),[42] advanced

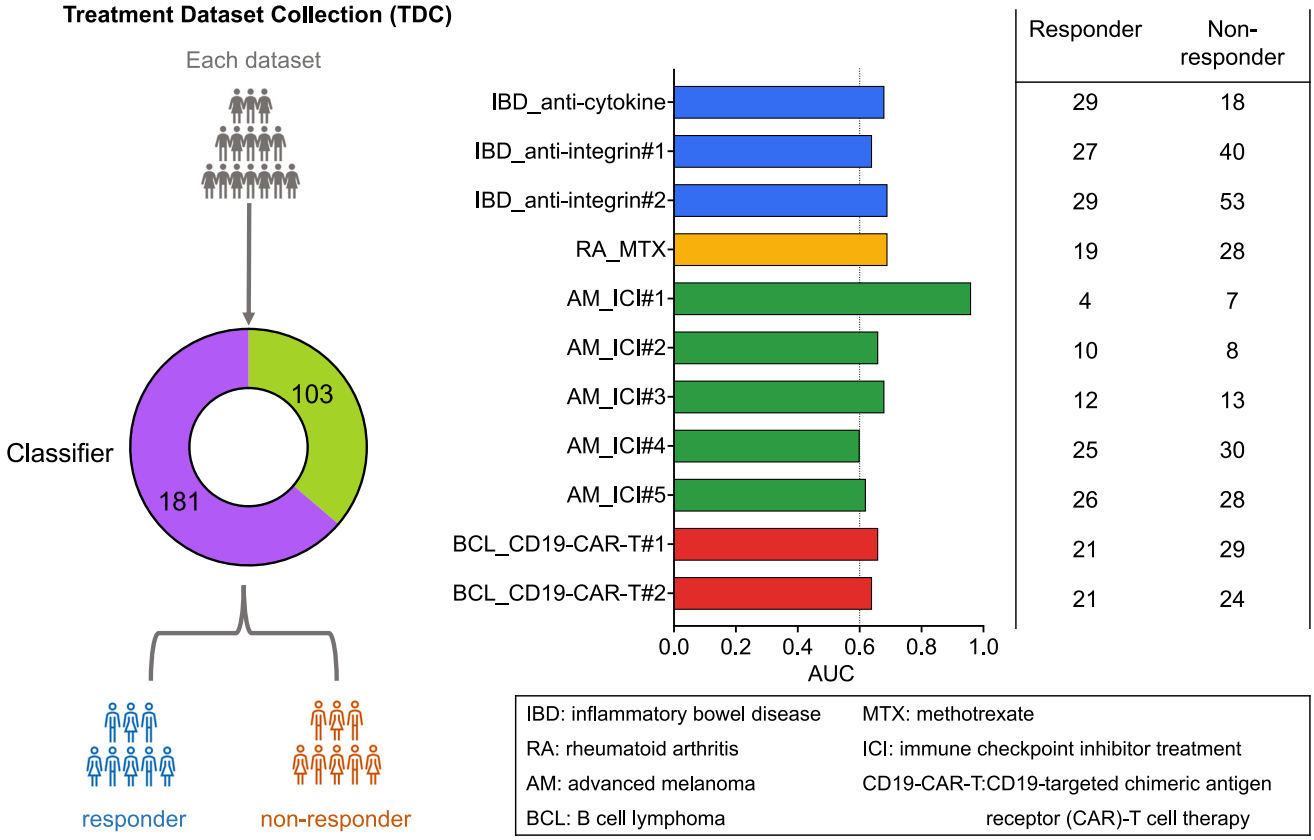


Figure 7. The CC-TCGs predict treatment outcomes in independent datasets

CC-TCG was used as a predictor in the treatment dataset collection to predict whether respondents (R) and non-responders (NR) were under treatment. For inflammatory bowel disease (IBD),14-week remission was used to determine the R and NR of patients with IBD to anti-cytokine or anti-antigen treatment. For rheumatoid arthritis (RA), responder to methotrexate (MTX) was defined a priori as any patient with new-onset RA with an improvement in the disease activity score in 28 joints (DAS28) of $\geq$1.8 by month four after initiation of MTX monotherapy. For advanced melanoma (AM), progression-free survival was used to determine R and NR to immune checkpoint inhibitor (ICI) treatment. For B cell lymphoma, the treating physician classified tumor response to CAR-T cell immunotherapy as complete or non-complete (partial remission, stable disease, progressive disease, or death) at 180 days after CAR-T cell infusion. The model was trained on the German cohort and validated by the US cohort. A sample size of R and NR is listed.
See also Figure S4.

melanoma,[43] and B cell lymphoma.[44] These datasets included responder and non-responder categories and are collectively referred to as the treatment dataset collection (TDC) (Figure 7). Using the 284 HQMAGs from the CC-TCG as reference genomes, we performed read recruitment analysis on the TDC metagenomic datasets. On average, we found that 16.22% and 13.06% of reads were recruited in the responder and non-responder samples, respectively. For each dataset in the TDC, a random forest classifier was built based on the abundance of the 284 HQMAGs in CC-TCG to predict therapeutic responses (Figure 7).

In the three IBD treatment datasets, CC-TCG predicted remission at week 14 with AUC values of 0.68 for IBD_anti-cytokine, 0.64 for IBD_anti-integrin#1, and 0.69 for IBD_anti-integrin#2 (Figures 7 and S4A). In the methotrexate treatment dataset for new-onset RA, CC-TCG predicted improvement with an AUC of 0.69 (Figures 7 and S4B).

The immune checkpoint inhibitor (ICI) treatment datasets for advanced melanoma included multiple cohorts, including Barcelona (AM_ICI#1), Leeds (AM_ICI#2), Manchester (AM_ICI#3), PRIMM-UK (AM_ICI#4), and PRIMM-NL (AM_ICI#5). On average, CC-TCG predicted overall response rate (ORR) and progression-free survival at 12 months (FPS12) with AUC values of 0.61 and 0.7 within each cohort, respectively (Figures 7 and S4C). The transportability of the prediction model from one cohort to another was insufficient. However, when pooling training datasets, as done in the LODO analysis, the prediction performance improved, reaching average AUC values of 0.66 and 0.67 for ORR and FPS12, respectively. In the datasets for CD19-CAR-T immunotherapy on B cell lymphoma, we trained a prediction model in the cohort from Germany (BCL_CD19-CAR-T#1) and validated it in the cohort from the United States (BCL_CD19-CAR-T#2). CC-TCG predicted responses to CD19-CAR-T immunotherapy with an AUC of 0.66 in the German cohort, and the model was sufficiently transportable to predict responses in the US cohort with an AUC of 0.64 (Figures 7 and S4D). These results highlight the potential of using CC-TCG to predict treatment effects across various diseases and therapies.

The dichotomy of conserved health-relevant functionalities of the genomes in the TCGs

Using HQMAGs allows us to annotate and understand the health-relevant functionalities of the two guilds in the TCGs. We noticed that with a 99% genomic ANI cutoff, the HQMAGs from the eight different TCGs found in QD and CCDC-I displayed limited overlap, with 701 of the 788 genomes in C-TCG being specific to just one TCG. At the same time, only 87 were present in more than one TCG. While this revealed the varying genomic composition of the HQMAGs from different TCGs derived from datasets of different studies, we observed similar functional profiles within C1As and C1Bs and the functional dichotomy between the two guilds across the 8 TCGs through non-targeted functional annotation of KEGG Ortholog (KO) in each guild. Pearson correlation analysis showed a strong correlation of KO profiles within C1As ($r = 0.89 \pm 0.071$, mean ± SD) and within C1Bs ($r = 0.96 \pm 0.023$). A lower correlation was observed between the C1As and C1Bs ($r = 0.76 \pm 0.093$). Principal coordinate analysis and clustering analysis, performed on Euclidean distance calculated from the KO profiles, further illustrated clear distinctions between the C1As and C1Bs from different TCGs (Figure S5), with a significant PERMANOVA test ($p = 0.002$). Among all the 5,870 identified KOs, 2,473 had significant differences in copy numbers between the two guilds, 354 were higher in C1As, and 2,119 were higher in C1Bs (Table S3). Thus, though the HQMAGs from different TCGs had few overlaps, the functions of each guild in different TCGs were highly consistent.

To understand why C1A in QD-TCG was promoted by the fibers and negatively correlated with HbA1c, we first sought to identify the carbohydrate-active enzyme (CAZy)-encoding genes and genes encoding key enzymes in short-chain fatty acid (SCFA) production to compare the genetic capacity for carbohydrate utilization between the two guilds. Within QD-TCG, compared with genomes in C1B, those in C1A were enriched with CAZy genes for arabinoxylan ($p < 0.001$), cellulose ($p < 0.001$) and had a lower proportion of CAZy genes for inulin utilization ($p < 0.01$) (Figure S6A). The two guilds were not different in genes for starch, pectin, and mucin utilization. Our previous study showed that gut microbiota benefited patients with T2D via acetic and butyric acid production from carbohydrate fermentation. Within QD-TCG, among the terminal genes for the butyrate biosynthetic pathways from both carbohydrates (i.e., but and buk) and proteins (i.e., atoA/D and 4Hbt), the copy number of *but* was significantly higher in C1A, and there was no difference in the other terminal genes between the two guilds (Figure S6B). More than one-third of the genomes in C1A harbored but gene, while less than 5% of the genomes in C1B had this gene (Fisher's exact test $p < 0.001$). Compared with C1B, C1A also trended higher in its genetic capacity for acetate production ($p = 0.06$) but had a lower genetic capacity for propionate production ($p < 0.05$) (Figure S6B). Thus, C1A in QD-TCG had a significantly higher genetic capacity for utilizing complex plant polysaccharides and producing butyrate and a lower genetic capacity for producing propionate. When we explored CAZy and SCFA-related genes in all eight different TCGs, this functional genetic distinction between C1A and C1B was further validated (Figures S6C and S6D). Moreover, CC-TCG also inherited this functional dichotomy (Figures S6E and S6F).

Next, to investigate the functional basis leading to the potential detrimental effect of C1B on the host, we identified antibiotic resistance genes (ARGs) and virulence factor genes (VFs) in the genomes of each guild. Within QD-TCG, C1A had 8 VF genes involved in 1 VF class and 1 ARG related to 1 antibiotic class, while C1B had 753 VF genes from 11 classes and 25 ARGs related to 7 antibiotic classes (Figure S7). In addition to QD-TCG, across all TCGs from CCDC-I, C1As consistently carried fewer VF genes and ARGs than C1Bs (Figure S7). CC-TCG inherited such a genetically functional difference between the two guilds, as evidenced by its C1A harboring 30 VF genes involved in 6 VF classes and 42 ARGs related to 5 antibiotic classes. At the same time, its C1B had 193 VF genes from 11 VF classes and 122 ARGs related to 7 antibiotic classes. In conclusion, these results show that the two guilds in TCGs have distinct genetic capacities, with C1A potentially beneficial and C1B detrimental.

DISCUSSION

Our study has uncovered a core microbiome signature marked by a dynamic, seesaw-like interplay between two competing bacterial guilds. This microbiome signature correlates with various host phenotypes, notably in individuals with T2D, emerging from our genome-specific, database-independent, and interaction-focused analytical method. Furthermore, this distinctive microbiome signature underpins our machine learning models, successfully differentiating between cases and controls across various conditions and predicting immunotherapy outcomes, underscoring its broad applicability. The articulation of the TCG model as a core microbiome signature represents a novel insight into our understanding of the microbiome-health nexus.

Based on the three distinct methodological pillars for our microbiome analysis, namely genome-specific analysis, database independence, and a focus on microbial interactions, this study concentrated on stable interactions within the gut microbiome. Such a stable interaction-focused approach has enhanced our understanding of the fundamental structure of the gut microbiome and devised a method to streamline the complex data inherent to this field. This approach allows us to maintain the integrity of crucial biological patterns while managing the high dimensionality and sparsity challenges of extensive microbiome datasets.

Our research encompassed approximately 4,000 metagenomic samples from 38 studies, covering 15 diseases over a decade across three continents. We pinpointed 284 HQMAGs within the CC-TCG from a pool of 8,000 by specifically targeting stably correlated genomes. Intriguingly, the refined predictive prowess of our TCG-informed models, when benchmarked against species-level, entire-genome-list, differential-feature, or high prevalent-feature models, substantiates the merit of our targeted approach (Data S3, pages 1–5). Notably, although high prevalence has been widely used as a criterion for core microbiome,[45] our results (Data S3, page 6) indicate that the prevalent gut microbiota members are not equally important to the host. In addition to high prevalence, stable interaction is critical in defining core health-relevant microbiomes. Despite selecting

a narrower array of microbial actors, the TCG models maintain, if not enhance, their predictive accuracy, highlighting the TCGs as parsimonious yet potent representatives of the microbiome's functional core.

In addition, our classification and prediction models based on TCGs showed consistent efficacy across different datasets and conditions, even without batch effect adjustments, demonstrating the robustness of our approach. Using stably correlated genomes has emerged as a powerful strategy to mitigate confounding factors like batch effects and technological discrepancies across studies. These results underscore the potential of stably correlated HQMAGs as a more precise biomarker in disease classification and prediction, setting a new standard for microbiome research.

The identified TCGs embody a functional dichotomy directly linked to host health, which holds promise but requires cautious interpretation across various datasets. In this binary, the beneficial guild (C1As) comprises genomes rich in CAZy genes essential for the digestion of dietary fiber and genes instrumental in the production of SCFAs, such as butyrate, which play vital roles in nutrition, metabolism, immune function, and overall physiology.[46] By contrast, what might be considered the detrimental counterpart (C1Bs) is characterized by genomes abundant with genes that confer antibiotic resistance and express VFs that may predispose to a pathogenic interaction with the host. It is thought that these "safe pathogens" or pathobionts might play a pivotal role in the developmental shaping of the immune system in early life.[47] However, the dominance of this guild over the beneficial one could contribute to a pro-inflammatory state, potentially underpinning various chronic conditions and the aging process. This observed functional dichotomy offers an intriguing lens to view and redefine eubiosis and dysbiosis, pointing to a delicate equilibrium within TCGs and their broader health implications.

Despite the stability of the TCG framework, the balance between these two guilds appears responsive to external influences like diet, as evidenced in our QD trial with individuals with T2D. In our QD trial, modifications in fiber intake were shown to alter the guild dynamics within the TCG framework and selectively foster the beneficial guild. These findings indicate that the proportional representation of the TCGs is not merely correlative but may be causally connected to metabolic health. The beneficial guild members ferment dietary fibers to extract energy to support their growth. This guild shows greater resilience in the more acidic gut environment caused by SCFA accumulation, a byproduct of their fiber digestion, indicating that dietary fibers serve not just as an energy source but also promote the growth of these beneficial microbes. They may progressively outcompete pathobionts due to the acidified environment,[48] SCFA's antimicrobial properties, and competitive exclusion. The beneficial guild thus may function similarly to "foundation species" in ecology, forming a supportive microbiome structure that underpins host health while making the gut environment inhibitive to pathobionts. We designate the beneficial guild (C1As) as the "foundation guild," postulating that their role may be analogous to towering trees within a forest ecosystem.[49] While drawing parallels between this foundation guild and ecological concepts offers an alluring analogy, we consider this a working hypothesis that merits further empirical exploration to solidify its role in the human gut microbiome.

The pervasive presence of the TCG structure across the case-control datasets reinforces the hypothesis that dietary fiber plays a pivotal role in shaping the backbone of the human gut microbiome throughout evolutionary trajectories. Indeed, studies on coprolites, the fossilized feces of ancient humans, have estimated a daily fiber intake of over 100 grams.[50,51] This substantial fiber intake experienced a dramatic decline only in the past century or so, and during this transition, the rampant emergence of pathobionts and inflammation is associated with nearly all major chronic conditions.[52,53] This might reflect the dwindling foundation guild within the TCGs, a repercussion of the steep reduction in dietary fiber intake to less than a tenth of what our ancestors consumed. Thus, dietary fiber may act as a constant energy infusion, nurturing an orderly TCG structure predominated by the foundation guild and bolstering host health within the intricate ecosystem of the gut microbiome. A broad spectrum of epidemiological and interventional studies has validated that dietary fibers can potentially deter and alleviate various disease conditions.[54,55] Our TCG model holds promise in unraveling the mechanisms underpinning this phenomenon.

We propose that members of our TCG model can be viewed as housekeeping microbes in the human gut microbiome, analogous to housekeeping genes in genomes.[56] Just as housekeeping genes are essential for the maintenance of basic cellular functions and are ubiquitously expressed in all cells of an organism, the core members identified in our TCG model represent foundational components of the gut microbiome that are critical for maintaining its overall stability and functionality. These housekeeping microbes could be involved in fundamental processes such as nutrient cycling, metabolic regulation, and immune modulation, similar to how housekeeping genes are involved in basic cellular processes such as transcription, translation, and replication.[56]

By focusing on these stable and core interactions, our TCG model aims to identify the most crucial microbial relationships that persist across different conditions and host environments. This approach allows us to pinpoint the key players in the gut ecosystem that support the broader microbiome structure and functionality, much like housekeeping genes support the basic infrastructure of the genome. Understanding these core microbial interactions provides valuable insights into the essential functions of the gut microbiome and its role in health and disease. It also offers a robust framework for developing targeted microbiome-based therapies, including a new wave of products like prebiotics, probiotics, synbiotics, postbiotics, microbiome transplantation, and live biotherapeutic products (LBPs)[57] focused on restoring and nurturing the foundation guild. Ultimately, the consistent core structure of TCGs that our research highlights may propel forward the fields of proactive and preventive healthcare, marking a significant stride in microbiome studies and their clinical ramifications.

Limitations of the study

Our approach encounters limitations when identifying disease-specific features within the gut microbiome. While insightful, the broad microbial signature we describe may overlook subtler,

disease-specific variations. This limitation is accentuated by the diversity of our datasets, which encompass multiple diseases, geographic regions, and ethnic groups, as well as the natural variability in microbiome composition among individuals.[58,59] Such variability could lead to missing critical components for understanding disease-specific microbial patterns. Future work should refine the resolution of disease-specific and personalized microbial features by integrating multi-omics data analysis and employing longitudinal designs that track microbiome evolution throughout disease across individuals. Additionally, our findings necessitate further exploration of high-order interactions[60] and the strength of pairwise correlations to enhance our understanding of microbial dynamics at a more granular level. Moreover, specific pathways enriched in the beneficial guild—particularly those related to fiber digestion and SCFA production—and the presence of VFs and ARGs in the unfavorable guild require more detailed mechanistic examination. These elements are pivotal for a comprehensive understanding of the microbial dynamics within these guilds but were beyond the scope of our study. Planned comprehensive bioinformatic analyses and experimental mechanistic studies will aim to fill these gaps. Lastly, further exploration of host-microbiome interactions within the TCG framework through controlled dietary and therapeutic interventions may provide valuable insights into the causative links between microbiome composition and disease, facilitating the development of more precise and effective microbiome-based diagnostics and therapies.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Liping Zhao (liping.zhao@rutgers.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

The metagenomic sequencing data have been deposited under accession numbers PRJEB15179 and PRJEB52732. The STAR Methods section shows the parameters of the bioinformatic tools applied in the study. Scripts and command lines related to this study are available from the lead contact (L.Z., liping.zhao@rutgers.edu) upon request.

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AUTHOR CONTRIBUTIONS

L.Z., Y.P., C.Z., and Yu Shi designed and supervised the study; T.X., F.Z., X.D., and Yu Shi managed the clinical research; T.X., X.D., D.W., J.F., Yajuan Shi, X.L., S.J., X.W., and F.Z. conducted the clinical study; T.X., X.W., H.F., and C.Z. performed clinical testing and data analysis; H.F. provided dietary counseling to the participants and conducted dietary analysis; T.X. and X.W. prepared the fecal DNA for sequencing. G.W., X.T., and C.Z. performed microbial data analysis of the QD trial. G.W. performed data validation in the independent datasets. L.Z., C.Z., G.W., N.Z., and Y.Y.L. wrote and revised the manuscript. G.W., T.X., N.Z., Y.Y.L., and X.D. contributed equally to this work.

DECLARATION OF INTERESTS

L.Z. is a co-founder of Notitia Biotechnologies Company. Patent WO2023212563A1 is related to this work. Patent app. no: 63/595,189 is related to this work.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited Data		
ENA	https://www.ebi.ac.uk/ena/browser/home	PRJEB15179, PRJEB52732
Download Data		
ENA	https://www.ebi.ac.uk/ena/browser/home	PRJDB4176 PRJEB10878 PRJEB12449 PRJEB13870 PRJEB21528 PRJEB21612 PRJEB27928 PRJEB28543 PRJEB29127 PRJEB29373 PRJEB38625 PRJEB43119 PRJEB43120 PRJEB43121 PRJEB43122 PRJEB43123 PRJEB43555 PRJEB54704 PRJEB6337 PRJEB7774 PRJNA375935 PRJNA384246 PRJNA400072 PRJNA422434 PRJNA431482 PRJNA433459 PRJNA447983 PRJNA638403 PRJNA682730 PRJNA685168 PRJNA686821 PRJNA832909
IBDMDB	https://ibdmdb.org/	HMP2
Software and Algorithms		
R project (version 4.2.1)	CRAN	https://cran.r-project.org/
caret (version 6.0-94) in R project	https://cran.r-project.org/web/packages/caret/index.html	N/A
vegan (version 2.6-4) in R project	https://cran.r-project.org/web/packages/vegan/index.html	N/A
KneadData	https://huttenhower.sph.harvard.edu/kneaddata/	N/A
Resistant gene identifier 5.2.1	https://card.mcmaster.ca/analyze/rgi	N/A
Prism version 9.5.1	GraphPad	https://www.graphpad.com/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Study design

The QD trial was conducted at the Qidong People's Hospital (Jiangsu, China) and examined the effect of a high fiber diet in free-living conditions in a cohort of individuals clinically diagnosed with T2DM. The study protocol was approved by the Ethics Committee of Shanghai General Hospital (2014KY104), and the study was conducted in accordance with the principles of the Declaration of

Helsinki. All participants provided written informed consent. The trial was registered with the Chinese Clinical Trial Registry (ChiCTR-IPC-14005346). The study design and participant flow are shown in Data S1, page 1.

T2DM patients of the Chinese Han ethnicity were recruited for the study (age: 37 - 70 years; HbA1c: 6.5%—12.0%). The Chinese Clinical Trial Registry (http://www.chictr.org.cn) provided a more detailed description of inclusion and exclusion criteria.

Patients received either a high fiber diet (WTP diet) as the treatment group (W group) or the usual care (Usual diet) as the control group (U group) for three months. Total caloric and macronutrient prescriptions were based on age-specific Chinese Dietary Reference Intakes (Chinese Nutrition Society, 2013). The WTP diet, based on whole grains, traditional Chinese medicinal foods, and prebiotics, included three ready-to-consume pre-prepared foods.[9] The usual care included standard dietary and exercise advice made according to the Chinese Diabetes Society guidelines for T2DM.[61] Patients in the W group were provided with the WTP diet to perform a self-administered intervention at home for three months, while patients in the U group accepted the usual care. W group stopped WTP diet intervention at the end of the third month (at M3). Then, W and U continued a one-year follow-up (M15). A meal-based food frequency questionnaire and 24-hour dietary recall were used to calculate nutrient intake based on the China Food Composition 2009.[62] Patients in both groups continued with their antidiabetic medications according to their physician prescriptions (Table S1).

Before a 2-week run-in period, all participants attended a lecture on diabetes intervention and improvements and received diabetes education and metabolic assessments. Based on the inclusion and exclusion criteria, 119 eligible individuals were enrolled and assigned into two groups in a 2:1 ratio (n = 79 in the W group, n = 40 in the U group) determined by SAS software.

Physical examinations were conducted at M0, M3, and M15 in Qidong People's Hospital (Jiangsu, China). Sample collection instructions were provided to the participants the day before. The participants provided the feces and first early morning urine as requested. After collecting the fasting venous blood sample, a 3-h meal tolerance test (Chinese buns containing 75 g of available carbohydrates; MTT test) was conducted, and the postprandial venous blood samples at 30, 60, 120, and 180 min were collected. All the blood samples were centrifuged at 3000 rpm for 20 min at 4°C after standing at room temperature for 30 min to obtain serum. The fasting blood serum was divided into two parts: one for hospital tests and the other for lab tests. The feces, urine, and serum samples were stored in dry ice immediately, then transported to the lab and frozen at -80°C. Subsequently, anthropometric markers and diabetic complication indexes were measured. Ewing test[63] and 24-hour dynamic electrocardiogram were conducted to estimate diabetic autonomic neuropathy (DAN). B-mode carotid ultrasound was performed to estimate atherosclerosis. Michigan Neuropathy Screening Instrument[64] was undertaken to estimate diabetic peripheral neuropathy (DPN). In addition, A meal-based food frequency questionnaire and the 24-hour dietary review were recorded for nutrient intake calculation. The drug use was self-reported and presented in Table S1.

The fasting venous blood was used to measure HbA1c, fasting blood glucose, fasting insulin, fasting C-peptide, C-reactive protein (CRP), blood routine examination, blood biochemical examination, and five analytes of the thyroid. At 30, 60, 120, and 180 min of MTT, the venous blood samples were used to measure the postprandial blood glucose, insulin, and C-peptide. Fasting early in the morning urine was used to calculate the routine urine examination and urinary microalbumin creatinine ratio. The measurements above were completed at Qidong People's Hospital. Fasting venous blood was used to quantify TNF-α (R&D Systems, MN, USA), lipopolysaccharide-binding protein (Hycult Biotech, PA, USA), leptin (P&C, PCDBH0287, China), and adiponectin (P&C, PCDBH0016, China) by enzyme-linked immunosorbent assays (ELISAs) at Shanghai Jiao Tong University.

The homeostatic model assessments of insulin resistance (HOMA-IR) and islet β-cell function (HOMA-β) were calculated based on fasting blood glucose (mmol/L) and fasting C-Peptide (pmol/L)[65] : HOMA-IR = 1.5 + FBG * Fasting-C-Peptide / 2800;

HOMA-β = 0.27 * Fasting-C-Peptide / (FBG - 3.5). Glomerular Filtration Rate was estimated by formula GFR (ml/min per 1.73 m^2) = 186 * $Scr^{-1.154}$ * $age^{-0.203}$ * 0.742 (if female) * 1.233 (if Chinese),[66] where Scr (serum creatinine) is in mg/dl and age is in years.

METHOD DETAILS

Metagenomic sequencing

DNA was extracted from fecal samples using the methods previously described.[8] Metagenomic sequencing was performed using Illumina Hiseq 3000 at GENEWIZ Co. (Beijing, China). Cluster generation, template hybridization, isothermal amplification, linearization, and blocking denaturing and hybridization of the sequencing primers were performed according to the workflow specified by the service provider. Libraries were constructed with an insert size of approximately 500 bp, followed by high-throughput sequencing to obtain paired-end reads with 150 bp in the forward and reverse directions. On average, 91.5 million ± 16.15 million (mean ± s.d.) reads per sample were obtained.

Data quality control

Prinseq[67] was used to: 1) trim the reads from the 3′ end until reaching the first nucleotide with a quality threshold of 20; 2) remove read pairs when either read was < 60 bp or contained "N" bases; and 3) de-duplicate the reads. Reads that could be aligned to the human genome (H. sapiens, UCSC hg19) were removed (aligned with Bowtie2[68] using –reorder –no-hd –no-contain –dovetail). On average, 86.48 million ± 15.12 million (mean ± s.d.) reads per sample were obtained for further analysis.

De novo assembly, abundance calculation, and taxonomic assignment of genomes

De novo assembly was performed for each sample using IDBA_UD[69] (–step 20 –mink 20 –maxk 100 –min_contig 500 –pre_correction). The assembled contigs were further binned using MetaBAT[70] (–minContig 1500 –superspecific -B 20). The quality of the bins was assessed using CheckM.[71] Bins had completeness > 95%, contamination < 5%, and strain heterogeneity < 5% were retained as high-quality draft genomes (Table S5). The assembled high-quality draft genomes were further dereplicated by using dRep.[72] DiTASiC,[73] which applied Kallisto for pseudo-alignment[74] and a generalized linear model for resolving shared reads among genomes, was used to calculate the abundance of the genomes in each sample, estimated counts with P-value > 0.05 were removed, and all samples were downsized to 36 million reads (One sample with read mapping ratio < 25%, which could not be well represented by the high-quality genomes, were removed in downstream analysis). Taxonomic assignment of the genomes was performed using GTDB-Tk[75] with default parameters (Table S5).

Gut microbiome network construction and analysis

In the W group, prevalent genomes shared by more than 75% of the samples at every time point were used to construct the co-abundance network at each time point. Fastspar,[14] a rapid and scalable correlation estimation tool for microbiome study, was used to calculate the correlations between the genomes with 1,000 permutations at each time point based on the abundances of the genomes across the patients, and the correlations with $P \leq 0.001$ were retained for further analysis. The networks were visualized with Cytoscape v3.8.1.[76] The layout of the nodes and edges was determined using the edge-weighted spring embedded layout and the correlation coefficient as weights. The links between the nodes are treated as metal springs attached to the pair of nodes. The correlation coefficient was used to determine the repulsion and attraction of the spring. The layout algorithm sets the position of the nodes to minimize the sum of forces in the network.

Identification of stable genome pairs and clusters

Let $N = \{1,2, \ldots, n\}$ be the finite set of genomes (HQMAGs). The networks are represented by an undirect graph, where the nodes represent the genomes and the edges strand for pairwise co-abundance correlations. Let $E = \{(i, j) \in N \times N | i < j\}$ be the set of edges. The requirement $i < j$ does not create asymmetries but is an easy way to have a unique label for each edge. A network on N corresponds to a vector $v \in \{-1, 0, 1\}^E$. For each edge (i,j), $v_{ij} = 1$ if i and j have a significant positive correlation, vij=-1 if i and j have a significant positive correlation, and $v_{ij}=0$ if there is no significant correlation between i and j. Let $G = \{E_1, E_2, \ldots E_k\}$ be the finite set of networks. If $\Sigma v_{ij}=k$ or $\Sigma v_{ij}=-k$, edge (i, j) was defined as a stable edge across the k networks. Nodes in the stable network were clustered using Connected Components Clustering analysis in Cystoscope. To identify if sub-clusters existed in Cluster C1, a clustering tree was based on the negative (set as -1) and positive correlations (set as 1) with the average linkage method followed by WGCNA analysis.[15]

Case-Control Dataset Collection I (CCDC-I)

Eleven independent metagenomic datasets on T2DM, LC, AS, ACVD, SCZ, CRC, and IBD were downloaded from the SRA or ENA database (Table S6). The group information, i.e., case or control, was collected from the corresponding papers or curatedMetagenomicData.[77] Quality control of raw reads was conducted by KneadData (https://bitbucket.org/biobakery/kneaddata). DiTASiC was used to recruit reads and estimate the abundance of the 141 HQMAGs in QD-CTG in each sample; estimated counts with P-value > 0.05 were removed and further converted to relative abundance divided by the total number of reads. A random forest classification model to classify case and control was constructed based on the estimated abundances of the HQMAGs in each dataset with leave-one-out cross-validation. High-quality reads were de novo assembled for each sample using MEGAHIT[78] (-min-contig-len 500, -presets meta-large). The assembled contigs were binned using MetaBAT 2[79] and MaxBin 2.[80] The quality of the bins was assessed using CheckM. Bins had completeness > 95%, contamination < 5%, and strain heterogeneity < 5% were retained as HQMAGs, which were further dereplicated by using dRep (Table S8). In the case and control groups, Fastspar with 1,000 permutations was used to calculate the co-abundance correlations based on the HQMAGs shared by more than 75% of the samples in both groups. Correlations with $P \leq 0.001$ were retained for further analysis. Robust, stable edges were defined as the unchanged positive/negative correlations between the same two genomes in both case and control groups. The same clustering analysis, performed in the QD trial, was conducted on the stable network. DiTASiC was used to estimate the abundance of HQMAGs in each TCG in each sample. A random forest classification model was constructed to classify case and control with leave-one-out cross-validation to test each TCG.

Identification of CC-TCG from C-TCG

We ranked the 788 HQMAGs in C-TCG based on their feature importance in the Random Forest models built for each CCDC-I dataset. Starting from the least important HQMAGs, we sequentially removed one HQMAG and trained new Random Forest models in each dataset, assessing AUC values for classifying cases and controls. This resulted in 788 different classifiers, each employing a different number of HQMAGs ranging from 1 to 788 in each CCDC-I dataset. We assigned ranks to HQMAG numbers based on the AUC values of their corresponding models, with lower ranks indicating higher AUC values. Classifiers trained on the top 302 HQMAGs showed the best classification performance. From the 302 HQMAGs, we further removed 18 that showed inconsistent C1A and C1B assignments across different TCGs.

Case-Control Dataset Collection II (CCDC-II)

Fifteen independent metagenomic datasets on AS, ASD, BD, COVID-19, CRC, GD, HT, MS, PC, and PD were downloaded from the SRA or ENA database (Table S6). The group information, i.e., case or control, was collected from the corresponding papers or curatedMetagenomicData. KneadData conducted quality control of raw reads. DiTASiC was used to recruit reads and estimate the abundance of HQMAGs in C-TCG and CC-TCG. A random forest classification model to classify case and control was constructed based on the estimated abundances of the HQMAGs in each dataset with leave-one-out cross-validation.

Treatment Dataset Collection (TDC)

Eleven independent metagenomic datasets on pre-treatment samples related to four diseases, including IBD, rheumatoid arthritis (RA), advanced melanoma, and B cell lymphoma, were downloaded from the SRA or ENA database (Table S6). Each sample's responder and non-responder categories were collected from the corresponding paper. Quality control of raw reads was conducted by KneadData. DiTASiC was used to recruit reads and estimate the abundance of HQMAGs in CC-TCG. A random forest classification model to predict responder and non-responder was constructed based on the estimated abundances of the HQMAGs in each dataset with leave-one-out cross-validation.

Gut microbiome functional analysis

Prokka[81] was used to annotate the HQMAGs. KEGG Orthologue (KO) IDs were assigned to the predicted protein sequences in each HQMAG by HMMSEARCH against KOfam using KofamKOALA with default parameters.[82] KOs were further assigned to KEGG modules. Antibiotic resistance genes were predicted using ResFinder[83] with default parameters. The identification of virulence factors was based on the core set of Virulence Factors of Pathogenic Bacteria Database (VFDB).[84] The predicted protein sequences were aligned to the reference sequence in VFDB using BLASTP (best hist with E-value < 1e-5, identity > 80%, and query coverage > 70%). Genes encoding carbohydrate-active enzymes (CAZys) were identified using dbCAN (releasee 6.0),[85] and the best-hit alignment was retained. Arabinoxylan-related CAZy families: CE1, CE2, CE4, CE6, CE7, GH10, GH11, GH115, GH43, GH51, GH67, GH3 and GH5; cellulose-related: GH1, GH44, GH48, GH8, GH9, GH3 and GH5; inulin-related: GH32 and GH91; mucin-related families: GH1, GH2, GH3, GH4, GH18, GH19, GH20, GH29, GH33, GH38, GH58, GH79, GH84, GH85, GH88, GH89, GH92, GH95, GH98, GH99, GH101, GH105, GH109, GH110, GH113, PL6, PL8, PL12, PL13 and PL21; pectin-related: CE12, CE8, GH28, PL1 and PL9; starch-related: GH13, GH31 and GH97. Genes encoding formate-tetrahydrofolate ligase, propionyl-CoA: succinate-CoA transferase, propionate CoA-transferase, 4Hbt, AtoA, AtoD, Buk and But were identified as described previously.[9]

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis was performed in the R environment (R version3.6.1). Friedman test, followed by the Nemenyi post-hoc test, was used for intra-group comparisons for repeat measurements. Mann-Whitney test (two-sided) was used to compare W and U at the same time point. Pearson Chi-square tests compared categorical data differences between groups or time points. PERMANOVA test (9,999 permutations) was used to compare the groups of gut microbiota structure.

The Mann-Whitney test (two-sided) and Fisher's exact test (two-sided) were used to compare the target functions between C1A and C1B. To compare HQMAGs in CC-TCG, hierarchical clustering analysis based on Jaccard distance on the KO profiles was conducted.

A linear mixed effect model with subject id as a random effect was applied to explore the associations between the abundance of guilds in QD-TCG and clinical parameters. For each HQMAG belonging to a guild, the robust clr-transformed abundance across each sample was first range-scaled. Subsequently, the guild abundance was obtained as the mean of the range-scaled abundances of HQMAGs belonging to that guild. The M0 and M3 time points were used to train the linear mixed effect model, and M15 was used for testing.

The Random Forest with leave-one-out cross-validation was used to perform classification analysis based on HQMAGs of TCGs in each CCDC-I, CCDC-II, and TDC dataset. The Random Forest with 10-fold cross-validation was used to perform classification analysis based on HQMAGs in CC-TCG for the universal model.

ADDITIONAL RESOURCES

The QD trial was registered with the Chinese Clinical Trial Registry (ChiCTR-IPC-14005346).

Supplemental figures

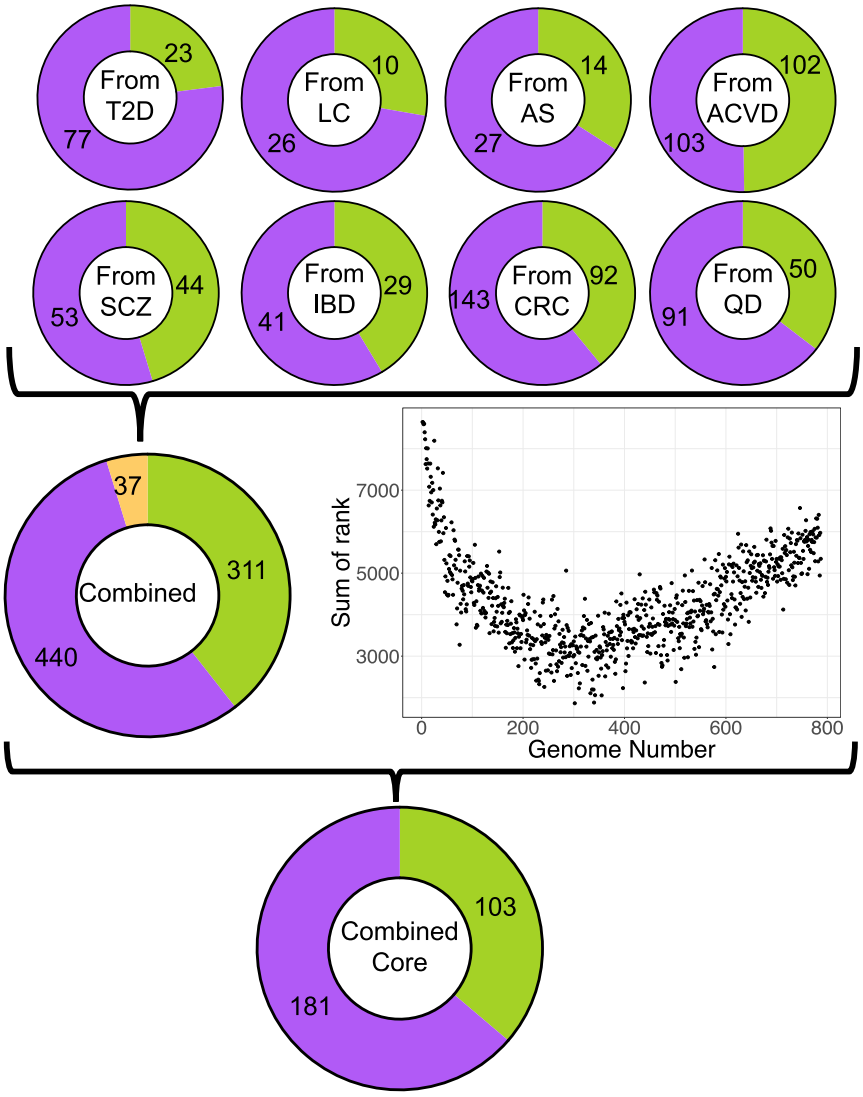


Figure S1. Identification of the combined and the combined core TCG from QD-TCG and TCGs from CCDC-I, related to Figure 1 and STAR Methods

All the HQMAGs in each set of TCG were dereplicated based on the cutoff of 99% average nucleotide identity (ANI) between two genomes. 788 non-redundant HQMAGs were obtained as the combined genomes of all eight sets of the TCGs. random forest classification model with leave-one-out cross-validation was constructed based on the 788 HQMAGs in each dataset. The HQMAGs were ranked based on their importance across all the models. From the least important HQMAGs (biggest importance rank), subsequently removing one HQMAG to do a random forest classification model in each dataset. In each dataset, the number of HQMAwasre ranked based on the area under the ROC curve (AUC) values. The scatter plot shows the relationship between HQMAG number and model performance. The y axis is the sum of rank based on AUC values (the smaller the value, the better the performance). The best performance was achieved at 302 HQMAGs as evidenced by the lowest sum of rank. After excluding 18 HQMAGs that exhibited inconsistent C1A and C1B assignments across the datasets, a total of 284 HQMAGs were kept from the 302 HQMAG as the combined core genomes of all the 8 sets of the TCGs.

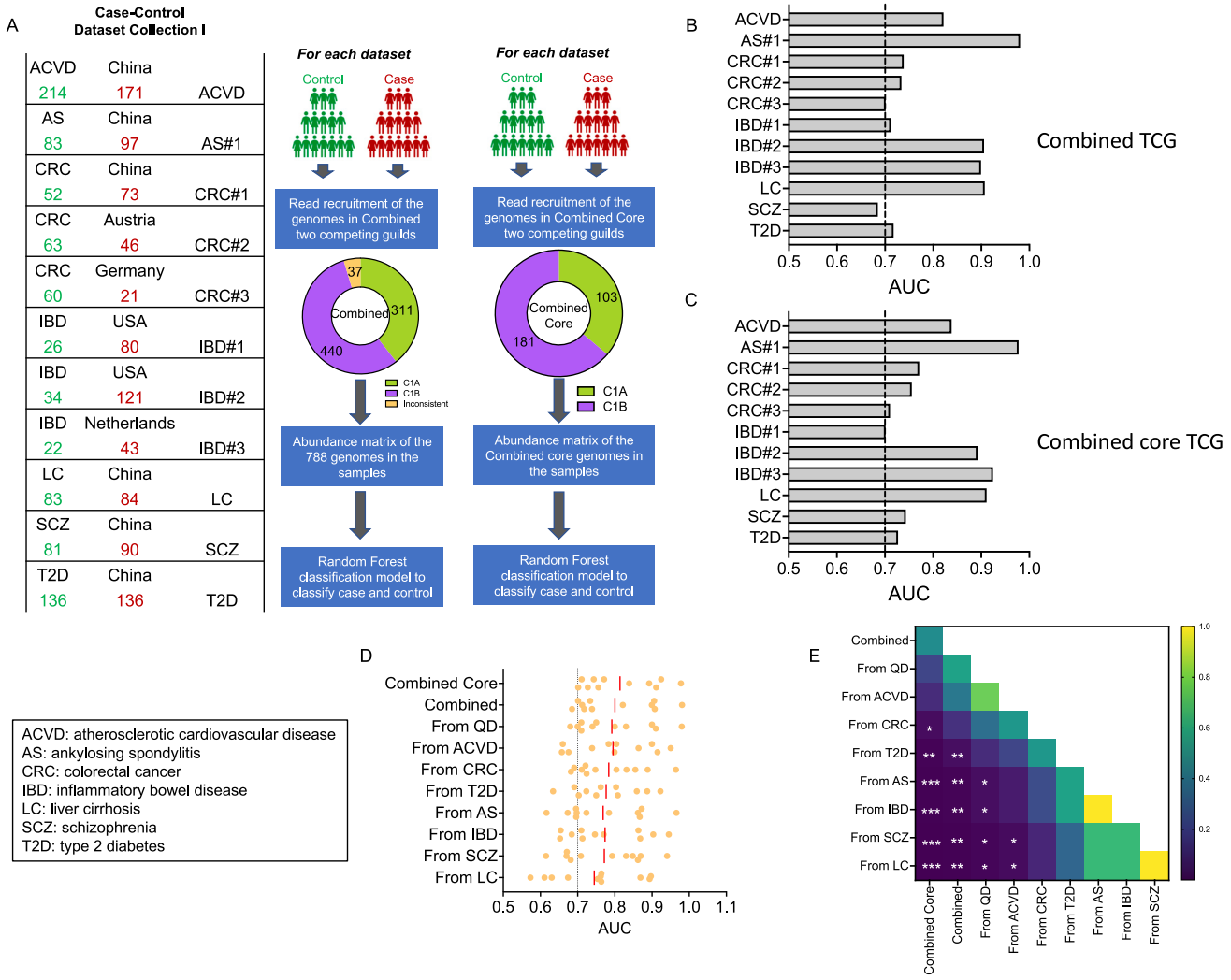


Figure S2. The combined and combined core genomes of all eight sets of the TCG support the classification of cases and controls in the case-control dataset collection I, related to Figure 1

(A) In each dataset in the collection, metagenomic reads were recruited to the combined genomes of 788 HQMAGs/combined core genome of 284 HQMAGs from all eight sets of the TCGs to estimate genome abundance in each sample.

(B) Based on the abundance matrix of the 788 HQMAGs of C-TCG, a random forest classification model with leave-one-out cross-validation was trained to classify case and control subjects in each dataset. The area under the ROC curves (AUC) is shown here.

(C) Based on the abundance matrix of the 284 HQMAGs of CC-TCG, a random forest classification model with leave-one-out cross-validation was trained to classify case and control subjects in each dataset, and the AUC values were shown here.

(D) The AUC values from Data S2 and (B) and (C) were shown as orange dots. Red line represents the mean AUC value. (D) Test the difference in classification performance between the different sets of the TCGs. AUC values in (C) were compared. Friedman test followed by Dunn's post hoc. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

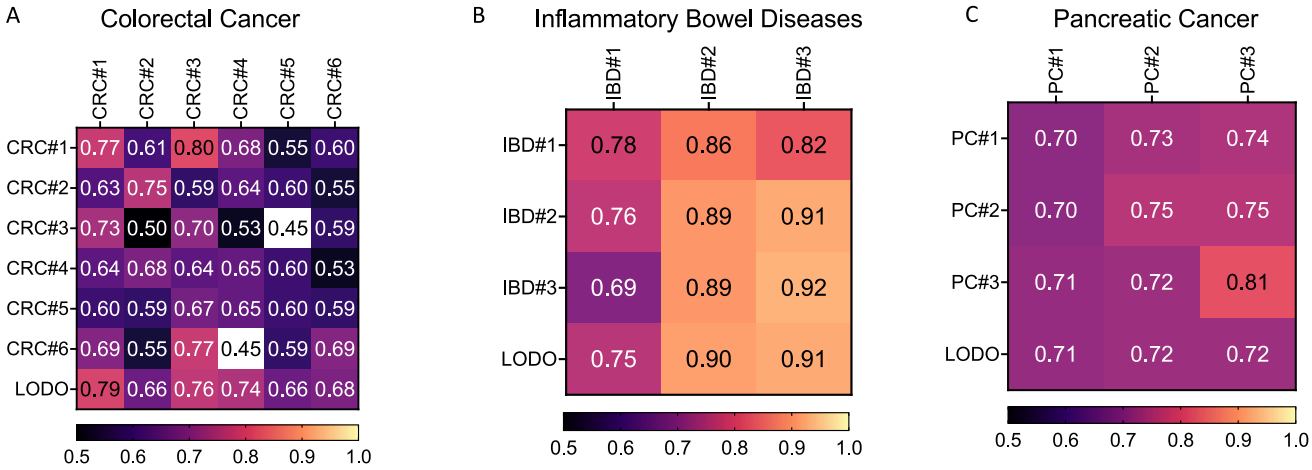


Figure S3. The discriminative power of the combined core genomes from all eight sets of the TCGs in classifying healthy individuals vs. patients across CRC, IBD, and PC datasets in the case-control dataset collection I and II, related to Figure 6

A prediction matrix was shown for the classification of cases and controls based on the combined core genomes from all eight sets of the TCGs within each dataset (diagonal values) across pairs of datasets (one dataset used for model training and the other for testing), and in a leave-one-dataset-out setting (training the model on all but one dataset and testing on the left-out dataset). Random forest classification model with leave-one-out cross-validation was applied. The area under the ROC curve (AUC) values were shown in the matrix.

(A) CRC. #1: case n = 74, control n = 54; #2: case n = 46, control n = 63; #3: case n = 22, control n = 60; #4: case n = 40, control n = 40; #5: case n = 61, control n = 52; and #6: case n = 52, control n = 52.

(B and C) (B) IBD. #1: case n = 80 control n = 26; #2: case n = 121, control n = 34; #3: case n = 43, control n = 22; (C) PC. #1: case n = 43 control n = 235; #2: case n = 57 control n = 50; and #3: case n = 44 control n = 32.

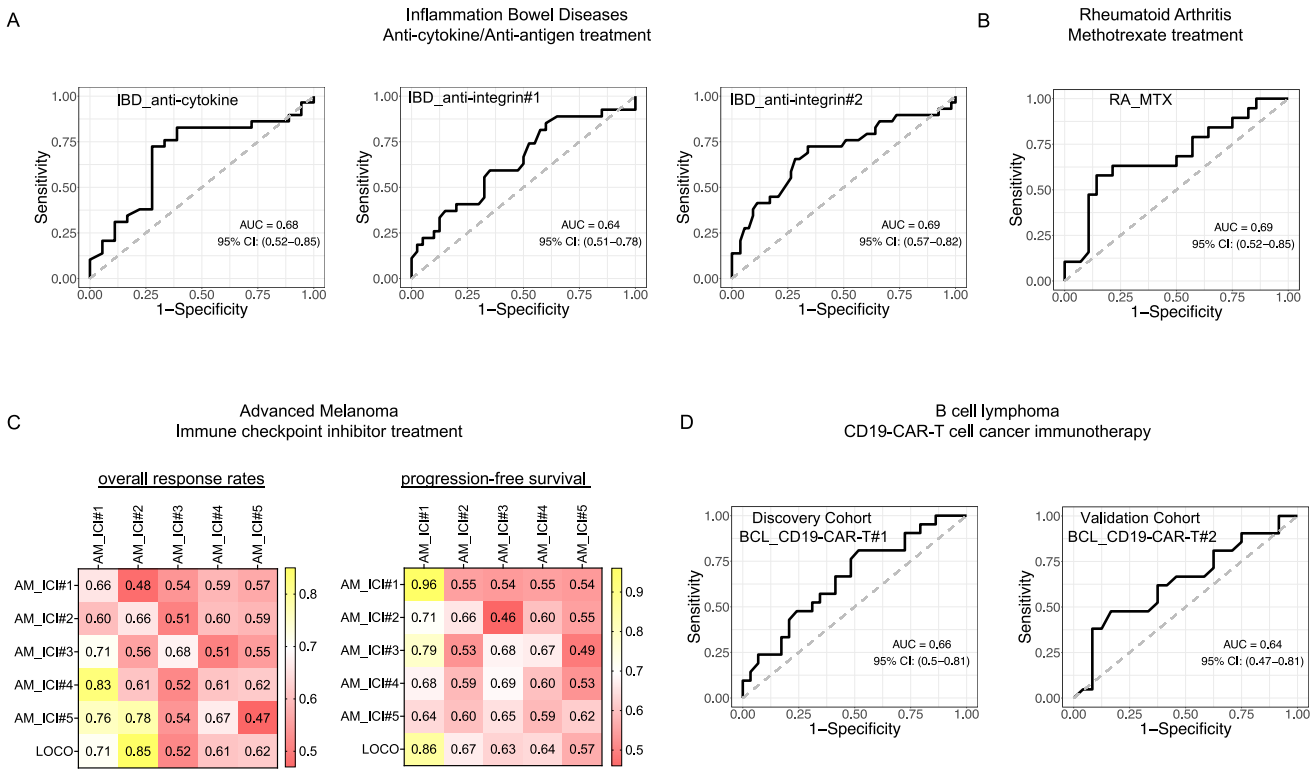


Figure S4. The combined core of the TCGs supports the prediction of therapeutic effects in the treatment dataset collection for inflammatory bowel disease, rheumatoid arthritis, advanced melanoma, and B cell lymphoma, related to Figure 7

The abundance of the combined core genomes (284 HQMAGs) in the pre-treatment samples was used as predictors in random forest classification models to predict responder (R) and non-responder (NR) under treatment. Area under the ROC curve (AUC) and AUC values were shown in the panels.

(A) 14-week remission was used to determine R and NR. IBD_anti-cytokine, *n* = 29 for R, *n* = 18 for NR; IBD_anti-integrin#1, R *n* = 27, NR *n* = 40; and IBD_anti-integrin#2, R *n* = 29, NR *n* = 53.

(B) Responder to MTX was defined a priori as any patient with new-onset RA with an improvement in the disease activity score in 28 joints (DAS28) (25) of ≥1.8 by month 4 after initiation of MTX monotherapy. R *n* = 19, NR *n* = 28.

(C) Overall response rate (ORR, left matrix) and progression-free survival (PFS12, right matrix) were used to determine R and NR, respectively. Prediction matrix for microbiome-based prediction of response assessed via ORR (left matrix) and PFS12 (right matrix) within each cohort (values on the diagonal) across pairs of cohorts (one cohort was used to train the model and the other for testing) and in the leave-one-cohort-out setting (training the model on all but one cohort and testing on the left-out cohort). ORR: R *n* = 94, NR *n* = 71; PFS12: R *n* = 77, NR *n* = 86.

(D) Tumor response to CAR-T cell immunotherapy was classified as either complete remission or non-complete remission (partial remission, stable disease, progressive disease, or death) at 180 days after CAR-T cell infusion by the treating physician. Model was trained on #1 (German cohort) and validated by #2 (US cohort). #1: R *n* = 21, NR *n* = 29; #2: R *n* = 21, NR *n* = 24.

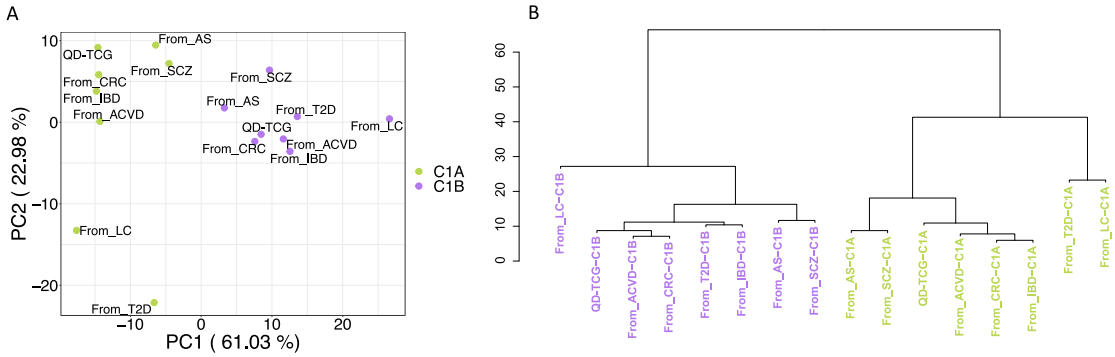


Figure S5. Guild-level functional consistency among the 8 TCGs found in the QD trial and CCDC-I, related to STAR Methods

(A) PCoA plot based on the Euclidean distance showed significant separations between C1As and C1Bs from the eight sets of TCGs.

(B) Ward linkage based on Euclidean distance showed separations between C1As and C1Bs from the eight sets of TCGs. Euclidean distance was calculated based on the mean copies of KEGG Orthology in each guild.

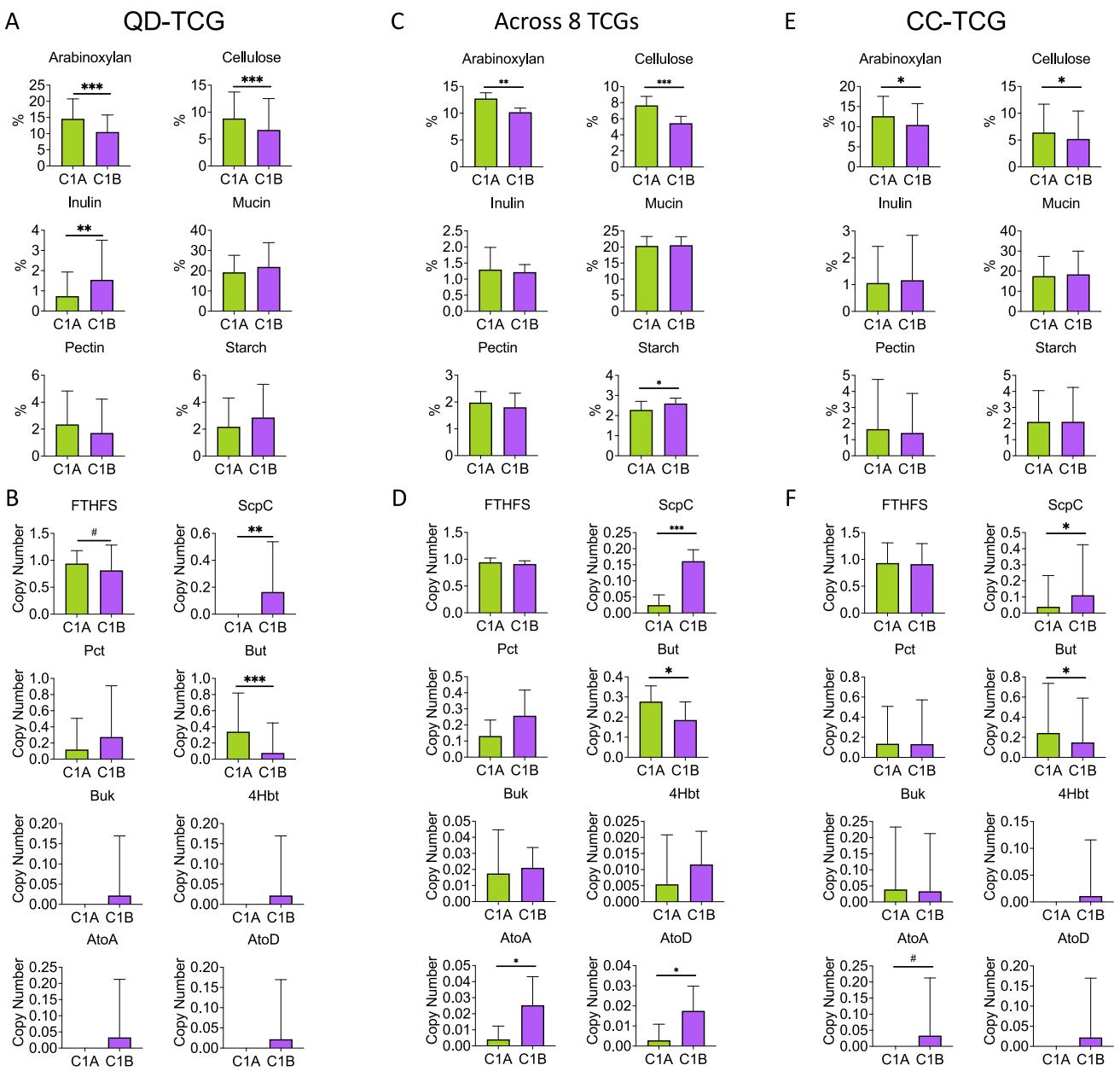


Figure S6. Difference in genes related to carbohydrate utilization and SCFA production between C1A and C1B, related to STAR Methods

(A, C, and E) Bar plot shows the proportion of CAZy genes for a particular substrate, calculated as the number of CAZy genes involved in its utilization divided by the total number of CAZy genes. Arabinoxylan-related CAZy families: CE1, CE2, CE4, CE6, CE7, GH10, GH11, GH115, GH43, GH51, GH67, GH3, and GH5; cellulose-related: GH1, GH44, GH48, GH8, GH9, GH3, and GH5; inulin-related: GH32 and GH91; mucin-related: GH1, GH2, GH3, GH4, GH18, GH19, GH20, GH29, GH33, GH38, GH58, GH79, GH84, GH85, GH88, GH89, GH92, GH95, GH98, GH99, GH101, GH105, GH109, GH110, GH113, PL6, PL8, PL12, PL13, and PL21; pectin-related: CE12, CE8, GH28, PL1, and PL9; and starch-related: GH13, GH31, and GH97.

(B, D, and F) Barplot showed the copy number of genes encoding FTHFS: formate-tetrahydrofolate ligase for acetate production; ScpC: propionyl-coenzyme A (CoA) succinate-CoA transferase and Pct: propionate-CoA transferase for propionate production; But, butyryl-coenzyme A (butyryl-CoA): acetate CoA transferase; Buk, butyrate kinase; 4Hbt, butyryl- CoA: 4-hydroxybutyrate CoA transferase; and Ato, butyryl-CoA:acetoacetate CoA transferase (AtoA: alpha subunit, AtoD: beta subunit) for butyrate production. Mann-Whitney test (two-sided) was used to analyze the difference between C1A 1 and C1B 2. # $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

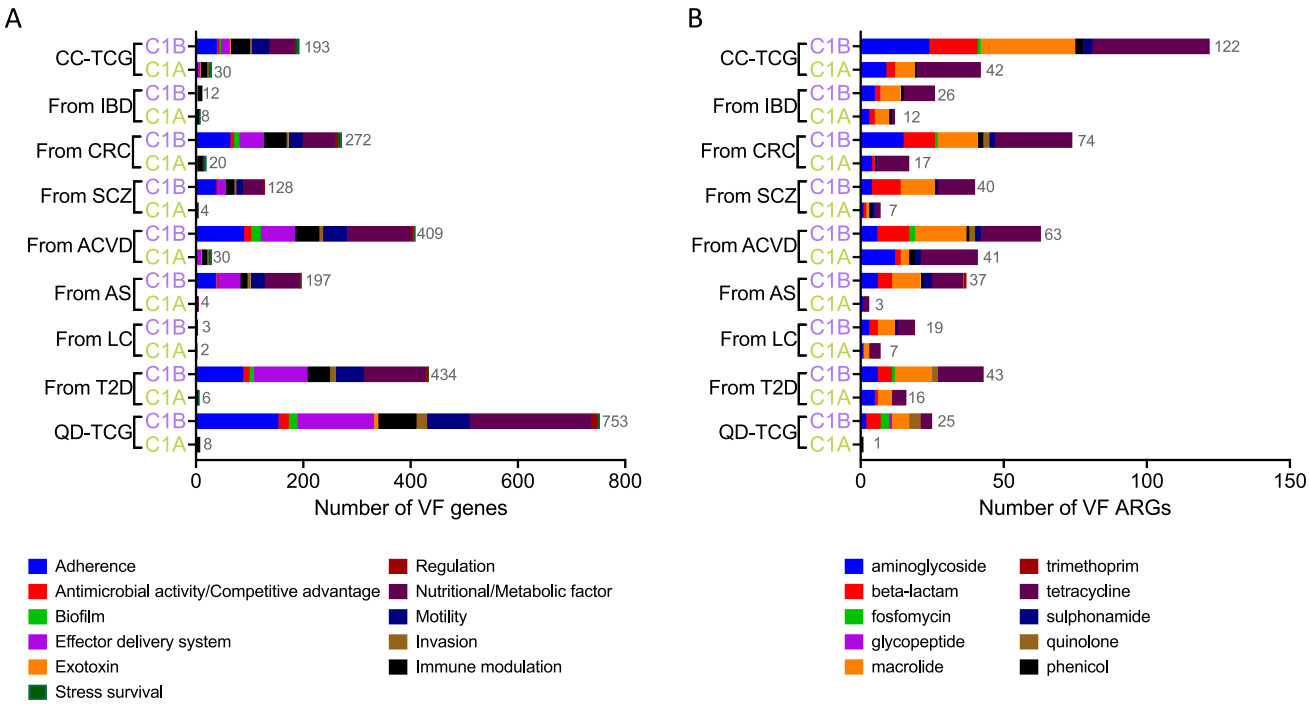


Figure S7. C1A consistently carried fewer VF genes and ARGs than C1B did, related to STAR Methods

(A) The stacked bar plot shows the number of VFs and the corresponding VF class.

(B) The stacked bar plot shows the number of ARGs and classes of antibiotics.